

Image Resampling Detection via Spectral Correlation with False Alarm Control

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March 19, 2025

Abstract

The detection of image resampling is a critical task in digital forensics, as it is essential for identifying manipulated media and verifying authenticity. Traditional approaches mostly rely on the correlations in the spatial domain caused by resampling, while the correlation in the Fourier domain related to the spectrum folding phenomenon during resampling is overlooked. In this work, we conduct a theoretical study of this phenomenon, and demonstrate that the spectral correlations caused by resampling are exploitable for image resampling detection. We propose an unsupervised method to detect anomalous spectral correlations, which consists of preprocessing an image to suppress natural correlations caused by the image content, and validating remaining significant detections using an *a contrario* framework. The proposed method is suitable for images with preserved aspect ratio, even if they are JPEG-compressed. Evaluated on resampled images with different anti-aliasing filters, scaling factors and JPEG compression levels, the proposed method demonstrates comprehensive superiority over the existing methods, and shows robustness to different interpolating filters and image sizes.

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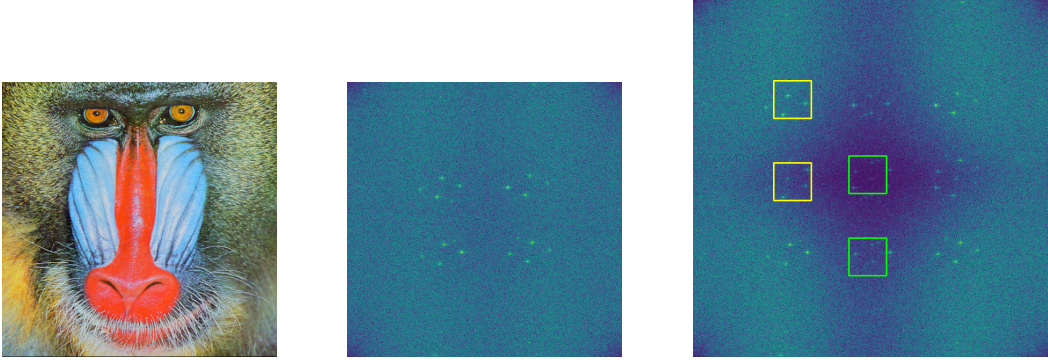


Fig. 1: We detect the anomalous correlations between small patch pairs in the spectrum of a resampled image. Left: the original “baboon” image in 512×512 . Middle: the spectrum of the original image preprocessed by the rank transform. Right: the spectrum of the upsampled image in 665×665 using bicubic interpolation. The upsampled image has also been preprocessed by the rank transform. In its spectrum, two pairs of patches separated by $d = (-512 \bmod 665) = 153$ components along the height are highly correlated. This leaves a hint for resampling detection.

Abstract—The detection of image resampling is a critical task in digital forensics, as it is essential for identifying manipulated media and verifying authenticity. Traditional approaches mostly rely on the correlations in the spatial domain caused by resampling, while the correlation in the Fourier domain related to the spectrum folding phenomenon during resampling is overlooked. In this work, we conduct a theoretical study of this phenomenon, and demonstrate that the spectral correlations caused by resampling are exploitable for image resampling detection. We propose an unsupervised method to detect anomalous spectral correlations, which consists of preprocessing an image to suppress natural correlations caused by the image content, and validating remaining significant detections using an *a contrario* framework. The proposed method is suitable for images with preserved aspect ratio, even if they are JPEG-compressed. Evaluated on resampled images with different anti-aliasing filters, scaling factors and JPEG compression levels, the proposed method demonstrates comprehensive superiority over the existing methods, and shows robustness to different interpolating filters and image sizes. Code is available at <https://github.com/li-yanhao/ird>.

Index Terms—image resampling detection, resizing detection, interpolation, image forensics, spectrum folding

I. INTRODUCTION

Image resampling is a common operation in digital image processing. It involves reconstructing the values on a new sampling grid from the pixels of the original grid using interpolation techniques, such as nearest-neighbor, bilinear, bicubic, or Lanczos interpolation. Most images are, for example, rescaled

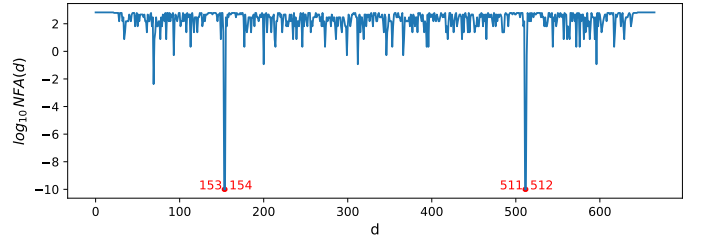


Fig. 2: The number of false alarms (NFAs) for all the possible distances d calculated on the upsampled “baboon” image in Fig. 1. Note that the height of the original image is $M_1 = 512$ and the height of the upsampled image is $N_1 = 655$. According to Sec. III-A, the significant NFAs should appear at distances $d = M_1(\bmod N_1) = 512$ and $d = -M_1(\bmod N_1) = 153$. The actual significant distances shown in the plot are slightly different from the theoretical ones due to the border effect of image resampling and of the preprocessing filter.

before being sent through social networks. Rescaling is also often involved in image forgery. The detection of resampling traces is therefore crucial to recover the editing history of an image, which is, for example, useful for downstream image authentication tasks [42].

Many forensic algorithms base their analysis on artefacts found early in the imaging pipeline [2, 4, 12, 20, 22–24, 38, 46], as such they can fail or can even yield false positives

when the image no longer has its original size. As shown in Karageorgiou et al. [26], AI-generated images are extremely difficult to detect when they have been resized, especially when using spectral artefacts that move with resampling [3, 31]. Knowing whether an image has been resampled is thus necessary to have a correct interpretation of these methods. Used locally, resampling detection could also be used by itself to detect local tampering, for instance if an object has been downsampled prior to splicing.

Beyond image forensics, many image processing steps modify the noise of an image, especially when they involve interpolation steps [5, 21, 25]. Resampling an image — and even more so upsampling it — similarly induces correlated noise in the image. Knowing that an image has been resampled, and ideally at which resampling factor, would be very useful information to pass to denoising algorithms when trying to restore images found in the wild, and would help select the right denoising method: as we go further from the pure Gaussian noise hypothesis under which most models are trained [48], other methods may perform better on resampled images.

Image resampling detection has been a subject of extensive research in the literature over the years. Its first prominent method dates back to Popescu and Farid [42], who applied the expectation-maximization (EM) algorithm to determine the correlation between neighboring pixels, resulting in a correlation probability map (p-map). The method is able to detect resampled images after a rotation, and was extended for forgery localization purposes by analyzing the incoherence of resampling traces across the image. Later, Kirchner [27] accelerated the computation of the p-map by replacing the time-consuming EM estimation of image filtering weights by a linear filter with preset coefficients. Furthermore, they presented an automatic approach to detect anomalies in the p-map, and studied the relationship between the geometric transformations of the image and the position of anomalous peaks in the p-map's spectrum. Thereafter, Kirchner et al. [28] studied how affine transformations of pre-compressed images affect p-map-based detectors and presented a variant of p-map to detect resampling on pre-compressed images. Nguyen and Katzenbeisser [39] also proposed improvements for p-map analysis [42] by applying the Radon transformation on a pseudo-pmap and looking for critical peaks in the transformed spectrum. As different resampling factors can lead to the same p-map and artefacts, it can be difficult to estimate an image's resampling factor. Pfennig and Kirchner [41] proposed to combine the analysis of p-maps with the spectral energy distribution to estimate the resampling factor of an image. Yet, image compression still introduces spurious spatial correlations which hinder the estimation.

Heuristic approaches other than p-map analysis were also proposed to investigate the local correlation in the spatial domain and reveal the periodic resampling artifacts in the Fourier domain. Gallagher [19] studied the hidden periodicity left by interpolation, and proposed to detect image resampling using the discrete Fourier transform (DFT) of the second difference. Following this idea, posterior works [7, 43] proposed to analyze the zero-crossings of the second difference of an image, and detect the image resampling in the DFT of the transformed image. Mahdian et al. [35] analyzed the hidden

periodicity left by interpolation and proposed to detect traces of an affine transformation. Specifically, the Radon transform is employed to detect the rotation traces. Then, resampling traces and noise analysis are combined for forgery detection [36]. Wang et al. [51] noticed that the local dependency between pixels induced by resampling leads to a lower rank of the matrix of a resampled image, and proposed to analyze the number of zero singular values of the singular value decomposition (SVD) for resampling detection. Building on this idea, Vázquez-Padín et al. [49] proposed a detection method that computes the SVD of a given image block and a measure of its degree of saturated pixels per row or per column, for discerning upsampled images from genuine ones. Vázquez-Padín et al. [50] presented an approach to design the optimal prefilter specific to a particular resampling factor and interpolation method for better estimating the resampling factor. Liu et al. [33] found that adjacent extrema intervals of a difference image obey a geometric distribution and that periodic peaks appear in the histogram of resampled JPEG images. They combined the characteristics of adjacent extrema intervals and spectrum artifacts for estimating downsampling factors on pre-JPEG-compressed images. Zhang et al. [54] exploited octant symmetric aliasing peaks (OSAP) caused by the coupling of resampling and post-JPEG compression to estimate resampling factors, and used an approximate OSAP model and a heuristic approach to enhance resampling features under low-quality JPEG compression.

Supervised learning-based methods have also been used to detect image resampling. A widely used approach is to leverage support vector machines (SVM) on hand-crafted features [17, 30, 40]. Methods based on deep neural networks have also been widely studied. Liang et al. [32] proposed a Convolutional Neural Network (CNN) to extract resampling feature patterns for detection. Bunk et al. [8] proposed a long-short-term memory (LSTM) network to detect and localize resampling traces. Flenner et al. [18] improved the method by Bunk et al. [8] by adopting the *a-contrario* framework to localize forged areas from resampling heat maps. Su et al. [47] modeled the resampled image as the result of a convolution between the original image and an unknown filter. They proposed to use a neural network for blind deconvolution, to simultaneously detect resampling and estimate the interpolation kernel. Lamba et al. [29] proposed an iterative pooling network and a branched network to tackle the resampling detection for images of varying dimensions. Cao et al. [9] presented a convolutional neural network to extract the resampling traces left in primary JPEG images followed by post-JPEG compression, and considered the resampling detection task as a multi-class classification problem. Bayar et al. [6] designed a constrained CNN to learn the low-level forensics traces and suppress image contents for detecting resampling in re-compressed images. Chen et al. [11] employed a densely connected CNN optimized for forensics to detect image resampling as well as other types of image manipulation. All of these methods show their effectiveness in resampling detection on several datasets. Nevertheless, supervised methods are not explainable, so no diagnosis is associated with a detection. Moreover, they suffer from domain-dependency and their generalization to new images with different statistics cannot

be guaranteed.

In this paper, we point out that instead of analyzing the correlations between pixels in the spatial domain like in [7, 19, 35, 44], the resampling detection problem can be addressed by investigating spectral correlations in the Fourier domain. It is well known that sampling a signal with a sampling rate inferior to the Nyquist frequency folds the frequency components above the Nyquist frequency onto the low frequencies and causes aliasing [37]. A similar phenomenon occurs in image resampling. When an image is downsampled, the high-frequency components can be folded onto the low frequencies unless a “perfect” anti-aliasing filter, removing all the aliased frequencies, is applied prior to resampling. A “perfect” anti-aliasing via a brutal cut-off of all the high frequencies above the Nyquist frequency leaves undesirable ringing effects in the edges of the image. A more practical anti-aliasing is to apply a smooth cut-off of the frequencies near the band-limit to balance the reduction between aliasing artifacts and ringing artifacts, which unavoidably leaves folding traces in the spectrum. As for upsampling, except for the upsampling by the sinc interpolation (which is equivalent to zero padding in the Fourier domain) we shall see that upsampling by any other interpolation method inevitably replicates the low frequencies to the high frequency band (see Fig. 1). In general, one can show that the resampling operation periodizes and folds the original spectrum to the destination spectrum, leaving anomalous correlations between the replicated frequency components in the destination spectrum.

We therefore propose to use the spectral correlation in a resampled image to detect anomalous correlations between spectrum patches. Furthermore, we shall see that the detected correlations are very unlikely in the residuals of non-resampled images. Hence, we can evaluate their significance using the *a contrario* statistical framework. This enables the computation of unsupervised detection thresholds. A non-linear preprocessing filter is adopted before analysis. It enables suppressing the image content, which may cause false positive detections, and uncovers subtle resampling traces. Extensive experiments show that the proposed resampling detection method is effective in different resampling scenarios and that it considerably outperforms all previous detection methods.

Our contributions are as follows:

- We conduct a theoretical study of the effect of signal resampling by interpolation in the Fourier domain. We specifically give general formulas revealing spectrum periodization and folding.
- We derive the existence of spectral correlations caused by the spectrum folding phenomenon occurring during resampling. Furthermore, we show that these are detectable and can be used as a trace of resampling.
- We present a method based on the *a contrario* framework that gives unsupervised detection thresholds to statistically validate the detected anomalous spectral correlations.
- A detailed comparison demonstrates that the method outperforms the compared methods by a large margin.

Symbol	Description
u	an original signal in the spatial domain
$\tilde{u}$	the DFT of an original signal
v	a resampled signal in the spatial domain
$\tilde{v}$	the DFT of a resampled signal
φ	an interpolating filter
$c_n(\varphi)$	the n -th Fourier coefficient of an interpolating filter φ
M	the size of an original signal
N	the size of a resampled signal

TABLE I: Table of notations used in Sec. II.

II. FOURIER ANALYSIS OF RESAMPLING

For the sake of clarity, we detail our notations for the discrete Fourier transform. Let us set $\omega_N := e^{\frac{i2\pi}{N}}$, $N \in \mathbb{R}_+$. The Discrete Fourier Transform (DFT) transforms a sequence of N samples $\{x_k\} := x_0, \dots, x_{N-1}$ into another sequence of N complex numbers $\{\tilde{x}_n\} := \tilde{x}_0, \dots, \tilde{x}_{N-1}$ by

$$\tilde{x}_n = \frac{1}{N} \sum_{k=0}^{N-1} x_k \cdot \omega_N^{-nk}, \quad n = 0, \dots, N-1. \quad (1)$$

The sequence $\{\tilde{x}_n\}$ defined above for $n = 0, \dots, N-1$ can be extended to $\mathbb{Z}$. The resulting sequence shall have a period of N since $\{\tilde{x}_n\}$ is N -periodic by definition. The inverse Discrete Fourier Transform (IDFT) transforms $\{\tilde{x}_n\}$ back to $\{x_k\}$ by

$$x_k = \sum_{n=0}^{N-1} \tilde{x}_n \cdot \omega_N^{kn}, \quad k = 0, \dots, N-1. \quad (2)$$

The Fourier series of an arbitrary real-valued T -periodic function $f \in L^2(0, T)$ is given by

$$f(x) = \sum_{n \in \mathbb{Z}} c_n(f) \cdot \omega_T^{nx}, \quad x \in \mathbb{R} \quad (3)$$

where $c_n(f) = \frac{1}{T} \int_0^T f(x) \cdot \omega_T^{-nx} dx$ are the Fourier coefficients associated to f . The common symbols in the following sections are summarized in Tab. I.

A. Spectrum folding formulas for resampling methods

Consider a 1D signal $\{u_k\} \in \mathbb{R}^M$ of M samples and its resampled version $\{v_k\} \in \mathbb{R}^N$ of N samples. Here $M, N \in \mathbb{N}^*$ can be any positive integers. All resampling methods proceed in two steps: first the samples $\{u_k\}$ are interpolated to a continuous function f using an interpolating filter φ , then f is resampled to the discrete signal $\{v_k\}$ of N samples:

$$\{u_k\} \in \mathbb{R}^M \xrightarrow[\varphi]{\text{interpolation}} f \xrightarrow{\text{resampling}} \{v_k\} \in \mathbb{R}^N.$$

The interpolation of $\{u_k\}$ to a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ is done by convolving $\{u_k\}$ with an interpolating filter φ :

$$f(x) = \sum_{k=0}^{M-1} u_k \cdot \varphi(x - k), \quad x \in \mathbb{R}. \quad (4)$$

Common interpolating filters φ include the nearest-neighbor, bilinear, bicubic and Lanczos filters. Their expressions and Fourier coefficients $c_n(\varphi)$ are summarized in Tab. II and visualized in Fig. 3.

Interpolator	Expression of $\varphi(x)$ defined on $[-\frac{M}{2}, \frac{M}{2}]$	Fourier coefficients $c_n(\varphi)$
Nearest neighbor	$\begin{cases} 1, & \text{if } x \leq \frac{1}{2} \\ 0, & \text{otherwise} \end{cases}$	$\frac{1}{M} \text{sinc}(\frac{n}{M})$
Linear	$\begin{cases} 1 - x , & \text{if } x \leq 1 \\ 0, & \text{otherwise} \end{cases}$	$\frac{1}{M} \text{sinc}^2(\frac{n}{M})$
Cubic	$\begin{cases} \frac{3}{2} x ^3 - \frac{5}{2}x^2 + 1, & \text{if } x \leq 1 \\ -\frac{1}{2} x ^3 + \frac{5}{2}x^2 - 4 x + 2, & \text{if } 1 < x \leq 2 \\ 0, & \text{otherwise} \end{cases}$	$\frac{1}{M} (3\text{sinc}^4(\frac{n}{M}) - 2\text{sinc}^2(\frac{n}{M}) \text{sinc}(\frac{2n}{M}))$
Lanczos	$\begin{cases} \text{sinc}(x) \text{sinc}(\frac{x}{3}), & \text{if } x \leq 3 \\ 0, & \text{otherwise} \end{cases}$	$\frac{18}{M} (\text{Rect} * \text{Rect}(3 \cdot) * \text{sinc}(6 \cdot))(\frac{n}{M}), \text{Rect}(\xi) := \begin{cases} 1, & \text{if } \xi \leq \frac{1}{2} \\ 0, & \text{otherwise} \end{cases}$

TABLE II: The expressions and the Fourier coefficients of different M -periodic interpolating filters φ . $*$ is the discrete convolution of two Fourier series.

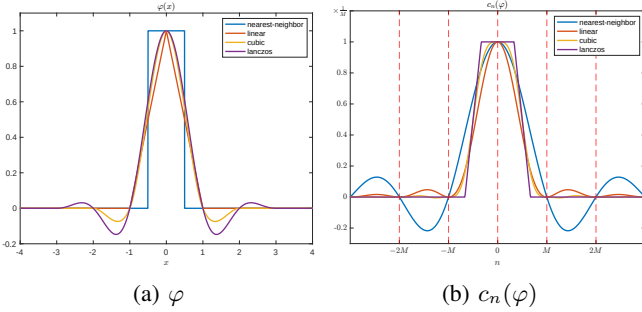


Fig. 3: The expressions of different interpolating filters φ and their Fourier coefficients $c_n(\varphi)$ listed in Tab. II.

Then the resampling from f to $\{v_k\}$ is achieved by evenly discretizing f on $[0, M]$ to N samples:

$$v_k = f\left(k \frac{M}{N}\right), \quad k = 0, \dots, N-1. \quad (5)$$

Theorem 1 (Modulated Spectrum Folding in 1D). *Suppose the interpolating filter φ is M -periodic. Denote the DFTs of $\{u_k\}$ and $\{v_k\}$ by $\{\tilde{u}_m\} \in \mathbb{C}^M$ and $\{\tilde{v}_n\} \in \mathbb{C}^N$, respectively. Denote the Fourier coefficients of φ and f by $\{c_p(\varphi)\}_{p \in \mathbb{Z}}$ and $\{c_p(f)\}_{p \in \mathbb{Z}}$, respectively. Then we have*

$$\tilde{v}_n = M \cdot \sum_{z \in \mathbb{Z}} c_{zN+n}(\varphi) \cdot \tilde{u}_{zN+n} \quad (6)$$

See Appendix A for the proof.

Theorem 1 describes the spectrum folding phenomenon in the Fourier domain when resampling a 1D signal of M samples into N samples with an interpolating filter. In simple words, it amounts to an N -periodization of the original M -periodic spectrum, modulated by the slow-varying Fourier coefficients of the interpolating filter.

The fact that the Fourier spectrum of the interpolating filter is slow-varying is essential, as it implies that one can observe near-replicas (up to a multiplicative constant) of the original spectrum in the spectrum of the interpolated image. The above theorem can be extended to any dimension. Here, we are particularly interested in 2D image resampling. Consider an image $\mathbf{u} \in \mathbb{R}^{M_1 \times M_2}$, resampled to an image $\mathbf{v} \in \mathbb{R}^{N_1 \times N_2}$ using an interpolating filter φ . Suppose that φ is (M_1, M_2) -periodic and that $\mathbf{u}$ is interpolated to an (M_1, M_2) -periodic

function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$. We obtain $\mathbf{v}$ by evenly sampling f on $[0, M_1] \times [0, M_2]$. The following Theorem again describes a spectrum folding phenomenon but in 2D.

Theorem 2 (Modulated Spectrum Folding in 2D). *Denote the DFTs of $\mathbf{u}$ and $\mathbf{v}$ by $\{\tilde{u}_{m_1, m_2}\} \in \mathbb{C}^{M_1 \times M_2}$ and $\{\tilde{v}_{n_1, n_2}\} \in \mathbb{C}^{N_1 \times N_2}$, respectively. Denote the Fourier coefficients of φ and f by $\{c_{p_1, p_2}(\varphi)\}_{(p_1, p_2) \in \mathbb{Z}^2}$ and $\{c_{p_1, p_2}(f)\}_{(p_1, p_2) \in \mathbb{Z}^2}$, respectively. Then we have*

$$\tilde{v}_{n_1, n_2} = M_1 M_2 \sum_{z_1, z_2 \in \mathbb{Z}} c_{z_1 N_1 + n_1, z_2 N_2 + n_2}(\varphi) \cdot \tilde{u}_{z_1 N_1 + n_1, z_2 N_2 + n_2} \quad (7)$$

See Appendix B for the proof for any dimension.

Theorem 2 again describes a spectrum folding phenomenon but in 2D.

B. Correlation between components in the spectrum

The spectrum folding Theorem states that a frequency component in the output spectrum is composed of several frequency components in the original spectrum. Conversely, we can show that a frequency component in the original spectrum contributes to different frequency components in the output spectrum, making these frequency components correlated. The distance between two correlated frequency components in the output spectrum depends on the original and output sizes.

For simplicity, let us again analyze the 1D case, in which an original signal is resampled from M to N samples by an interpolating filter φ . As before, we denote the DFT of the original signal by $\{\tilde{u}_n\} \in \mathbb{C}^M$ and the DFT of the resampled signal by $\{\tilde{v}_n\} \in \mathbb{C}^N$. Denote the Fourier coefficients of the interpolating filter φ by $\{c_p(\varphi)\}_{p \in \mathbb{Z}}$.

From Theorem 1 we know that two frequency components separated by kM components, $k \in \mathbb{Z}$ being an arbitrary integer, can be correlated. Indeed, for any $n \in \mathbb{Z}$,

$$\tilde{v}_n = M \sum_{z \in \mathbb{Z}} c_{zN+n}(\varphi) \cdot \tilde{u}_{zN+n} \quad (8)$$

and as $\{\tilde{u}_m\}$ is M -periodic,

$$\begin{aligned} \tilde{v}_{n+kM} &= M \sum_{z \in \mathbb{Z}} c_{zN+n+kM}(\varphi) \cdot \tilde{u}_{zN+n+kM} \\ &= M \sum_{z \in \mathbb{Z}} c_{zN+n+kM}(\varphi) \cdot \tilde{u}_{zN+n} \end{aligned} \quad (9)$$

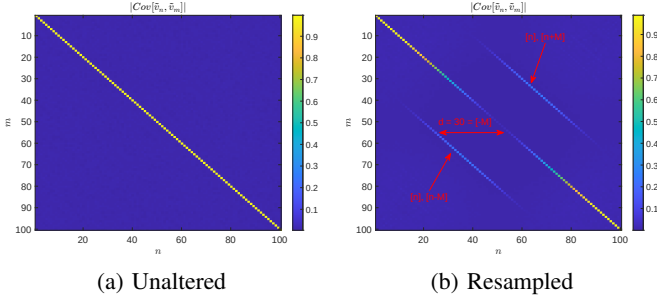


Fig. 4: $|\text{Cov}[\tilde{v}_n, \tilde{v}_m]|$ of the spectrum $\tilde{v}$ of (a) an unaltered white noise and (b) a resampled white noise. Both sequences have $N = 100$ samples. The original size of the resampled noise in (b) is $M = 70$. Note that two frequency components $\tilde{v}_n$ and $\tilde{v}_m$ in the spectrum of the resampled noise are likely to have strong covariance when $m = n \pm M(\text{mod } N)$. On the other hand, any two different frequency components in the unaltered noise are uncorrelated.

where we can see $\tilde{v}_n$ and $\tilde{v}_{n+kM}$ are both weighted sums of the same original frequency components $\{\tilde{u}_{zN+n}, z \in \mathbb{Z}\}$ with different weights from $\{c_p(\varphi)\}$. Hence, $\tilde{v}_n$ and $\tilde{v}_{n+kM}$ can be correlated due to the presence of the same $\{\tilde{u}_{zN+n}, z \in \mathbb{Z}\}$. We will study the properties of frequency correlation in the specific case of white noise, in particular, we shall see that:

- $\tilde{v}_n$ and $\tilde{v}_{n+kM}$ are likely to be correlated for $k \in \mathbb{Z}^*$;
- the correlation between $\tilde{v}_n$ and $\tilde{v}_{n+kM}$ is the strongest when $|k| = 1$ and vanishes as $|k|$ increases.

Fig. 4 shows the magnitude of the covariances of the DFTs $|\text{Cov}[\tilde{v}_n, \tilde{v}_m]|$ of an unaltered white noise and a resampled white noise, estimated from 10K simulations. Both noise sequences have $N = 100$ samples. The original size of the resampled white noise is $M = 70$. One can clearly see that two different frequency components $\tilde{v}_n$ and $\tilde{v}_m$ of the resampled noise are likely to be correlated when $m = n \pm M(\text{mod } N)$, and are almost uncorrelated for other component pairs. Indeed, $\text{Cov}[\tilde{v}_n, \tilde{v}_{n+kM}]$ is non-zero for any $k \in \mathbb{Z}$, but vanishes rapidly and is deemed negligible as $|k|$ increases. In contrast, any two different frequency components are uncorrelated for the unaltered white noise.

Theorem 3. *Let the original signal $\{u_k\} \in \mathbb{R}^M$ be a sequence of white noise with $\text{Var}[u_k] = \sigma^2$ and $\text{E}[u_k] = 0$, which is resampled to $\{v_k\} \in \mathbb{R}^N$ using an M -periodic interpolating filter $\varphi : \mathbb{R} \mapsto \mathbb{R}$. Let $\tilde{v}_m$ and $\tilde{v}_n$ be two arbitrary frequency components in the DFT of the resampled signal, $m, n \in \mathbb{Z}$. Denote by $L = \text{lcm}(M, N)$ and $\text{gcd}(M, N)$ the least common multiple and the greatest common divisor of M and N , respectively. We have $\text{Cov}[\tilde{v}_n, \tilde{v}_m]$ given by:*

- if $m - n$ is not a multiple of $\text{gcd}(M, N)$, $\text{Cov}[\tilde{v}_n, \tilde{v}_m] = 0$;
- if $m - n$ is a multiple of $\text{gcd}(M, N)$, define $\mathcal{K}(x)$ as the solution of

$$\begin{aligned} \min_{k'} & |k'| \\ \text{s.t.} & k'M(\text{mod } N) = x(\text{mod } N) \\ & k' \in \mathbb{Z} \end{aligned} \quad (10)$$

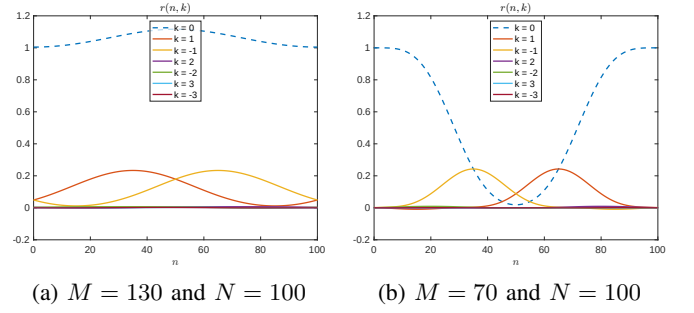


Fig. 5: $n \mapsto r(n, k)$ for different small values of k in Eq. 12 for downsampling (Fig. 5a) and upsampling (Fig. 5b). $r(n, k)$ is non-negligible only for $k = 0, \pm 1$ and vanishes rapidly as $|k|$ increases.

and let $k = \mathcal{K}(m - n)$, then $m(\text{mod } N) = n + kM(\text{mod } N)$ and the covariance is given by

$$\begin{aligned} \text{Cov}[\tilde{v}_n, \tilde{v}_m] &= \text{Cov}[\tilde{v}_n, \tilde{v}_{n+kM}] \\ &= M\sigma^2 \sum_{q \in \mathbb{Z}} \sum_{z \in \mathbb{Z}} c_{zN+n}(\varphi) \cdot \overline{c_{zN+n+kM+qL}(\varphi)} \end{aligned} \quad (11)$$

See Appendix C for the proof.

Theorem 3 describes the covariance between two frequency components $\tilde{v}_n$ and $\tilde{v}_m$ of a resampled white noise, $m - n$ being divisible by $\text{gcd}(M, N)$, as the sum of inner products of pairs of frequency components separated by $kM + qL$ components, where q traverses $\mathbb{Z}$ and $k = \mathcal{K}(m - n)$ is fixed. If we assume that the interpolating filter φ is a low-pass filter, its Fourier coefficients $c_p(\varphi)$ vanish as $|p|$ increases, and each inner product of Eq. 11 vanishes rapidly as the distance $|kM + qL|$ increases. The distance is independent of z and changes as q changes. Intuitively, we can expect the outer summation with index variable q to be dominated by the largest term associated to the dominant q of which the distance $|kM + qL|$ is minimal. Because the solutions of k' of Eq. 10 are $\mathbf{K} = \{k + q\frac{L}{M}, q \in \mathbb{Z}\}$ and by definition $|k| = \min_{k' \in \mathbf{K}} |k'|$, the minimal distance among $\{|kM + qL|, q \in \mathbb{Z}\}$ is $|kM|$. Therefore, the outer summation is dominated by the term at $q = 0$ and $\text{Cov}[\tilde{v}_n, \tilde{v}_{n+kM}] \approx M\sigma^2 \sum_{z \in \mathbb{Z}} c_{zN+n}(\varphi) \overline{c_{zN+n+kM}(\varphi)}$ where all the inner products share the common distance $|kM|$. More importantly, when n is fixed, if m changes so that $|k| = |\mathcal{K}(m - n)|$ increases, then the common distance $|kM|$ increases and $|\text{Cov}[\tilde{v}_n, \tilde{v}_m]|$ decreases.

To better illustrate the relationship between $|k|$ and $|\text{Cov}[\tilde{v}_n, \tilde{v}_{n+kM}]|$, we define

$$\begin{aligned} r(n, k) &:= \frac{\text{Cov}[\tilde{v}_n, \tilde{v}_{n+kM}]}{M\sigma^2} \\ &= \sum_{q \in \mathbb{Z}} \sum_{z \in \mathbb{Z}} c_{zN+n}(\varphi) \cdot \overline{c_{zN+n+kM+qL}(\varphi)} \end{aligned} \quad (12)$$

which is real-valued for a real-valued and even interpolating filter¹ φ , and plot the curves of $n \mapsto r(n, k)$ corresponding to the cubic filter in the case of downsampling ($M > N$) or upsampling ($M < N$) for different small values of k in Fig. 5.

¹This is due to the fact that the Fourier coefficients of a real-valued and even function are real-valued.

It can be seen that the $r(n, k)$ vanishes rapidly as $|k|$ increases, and is non-negligible only for $k = 0, \pm 1$. This confirms the observation in Fig. 4 that only the values on the upper and lower diagonals, $\{\text{Cov}[\tilde{v}_n, \tilde{v}_{n \pm M \pmod{N}}]\}_n$, are non-negligible. The non-negligible correlation between $\tilde{v}_n$ and $\tilde{v}_{n+kM}$ for small $|k|$, especially for $|k| = 1$, is an important trace left by the resampling operation and hence a hint for its detection.

C. Correlation between patches in the spectrum

Despite the non-negligible covariance between two frequency components $\tilde{v}_n$ and $\tilde{v}_{n+kM}$ for small $|k|$, in practice we only have access to one single sample of $\tilde{v}_n$ and $\tilde{v}_{n+kM}$ from the DFT of the signal in question, which makes the estimation of $\text{Cov}[\tilde{v}_n, \tilde{v}_{n+kM}]$ imprecise. We can instead measure the correlation between the means of two small segments of the spectrum separated by kM frequency components by varying n in a small interval. Concretely, let $\mathcal{I}_l = [-l, l]$ be a small interval with $l \ll M$ and let $\tilde{w}_n = \frac{1}{l} \sum_{\Delta \in \mathcal{I}_l} \tilde{v}_{n+\Delta}$ and $\tilde{w}_{n+kM} = \frac{1}{l} \sum_{\Delta \in \mathcal{I}_l} \tilde{v}_{n+kM+\Delta}$ be the means of two segments separated by kM components, then the correlation between $\tilde{w}_n$ and $\tilde{w}_{n+kM}$ is given by

$$\begin{aligned} \rho(\tilde{w}_n, \tilde{w}_{n+kM}) &= \frac{|\text{Cov}[\tilde{w}_n, \tilde{w}_{n+kM}]|}{\sqrt{|\text{Var}[\tilde{w}_n]|} \sqrt{|\text{Var}[\tilde{w}_{n+kM}]|}} \\ &\approx \frac{|\sum_{\Delta \in \mathcal{I}_l} \text{Cov}[\tilde{v}_{n+\Delta}, \tilde{v}_{n+kM+\Delta}]|}{\sqrt{|\sum_{\Delta \in \mathcal{I}_l} \text{Var}[\tilde{v}_{n+\Delta}]|} \sqrt{|\sum_{\Delta \in \mathcal{I}_l} \text{Var}[\tilde{v}_{n+kM+\Delta}]|}} \end{aligned} \quad (13)$$

which amounts to calculating the Pearson correlation coefficient (PCC) of the two segments $\{\tilde{v}_{n+\Delta}\}_{\Delta \in \mathcal{I}}$ and $\{\tilde{v}_{n+kM+\Delta}\}_{\Delta \in \mathcal{I}}$ in practice. $\text{Cov}[\tilde{w}_n, \tilde{w}_{n+kM}]$, $\text{Var}[\tilde{w}_n]$ and $\text{Var}[\tilde{w}_{n+kM}]$ in Eq. 13 is respectively expanded at the second equality as the sum of the covariances of components at distance kM and the variances of each of the component, because any other pairs of components with the two small segments separated at distances different from 0 or kM have negligible covariances.² Combining with Eq. 12, Eq. 13 can be further written as

$$\begin{aligned} \rho(\tilde{w}_n, \tilde{w}_{n+kM}) &= \frac{|\frac{1}{l} \sum_{\Delta \in \mathcal{I}_l} r(n+\Delta, k)|}{\sqrt{|\frac{1}{l} \sum_{\Delta \in \mathcal{I}_l} r(n+\Delta, 0)|} \sqrt{|\frac{1}{l} \sum_{\Delta \in \mathcal{I}_l} r(n+kM+\Delta, 0)|}} \end{aligned} \quad (14)$$

The denominator is bounded as $n \mapsto r(n, 0)$ is bounded, and the numerator as a moving average of $n \mapsto r(n, k)$ is non-negligible. Thus, $\rho(\tilde{w}_n, \tilde{w}_{n+kM})$ is non-negligible. $\tilde{w}_n$ and $\tilde{w}_{n+kM}$ being the means of multiple frequency components, the correlation $\rho(\tilde{w}_n, \tilde{w}_{n+kM})$ is observable in practice with less uncertainty. Also, the correlation eliminates the constant $M\sigma^2$ in the covariance and makes the measure invariant to the size and variance of the original signal.

Fig. 6. shows the correlation between two different segments $\rho(\tilde{w}_n, \tilde{w}_{n+kM})$ (Eq. 14) as a function of the offset n for

² $\text{Cov}[\tilde{w}_n, \tilde{w}_{n+kM}] = \frac{1}{l^2} (\sum_{\Delta \in \mathcal{I}_l} \text{Cov}[\tilde{v}_{n+\Delta}, \tilde{v}_{n+kM+\Delta}] + \sum_{\Delta_1, \Delta_2 \in \mathcal{I}_l, \Delta_1 \neq \Delta_2} \text{Cov}[\tilde{v}_{n+\Delta_1}, \tilde{v}_{n+kM+\Delta_2}])$. When $|k|$ is small and $\Delta_2 - \Delta_1$ is small but non-zero, $(n+kM+\Delta_2) - (n+\Delta_1) = kM+\Delta_2-\Delta_1$ is either not a multiple of $\text{gcd}(M, N)$ or related to $|K(kM+\Delta_2-\Delta_1)| \gg 1$, which results in negligible $\text{Cov}[\tilde{v}_{n+\Delta_1}, \tilde{v}_{n+kM+\Delta_2}]$.

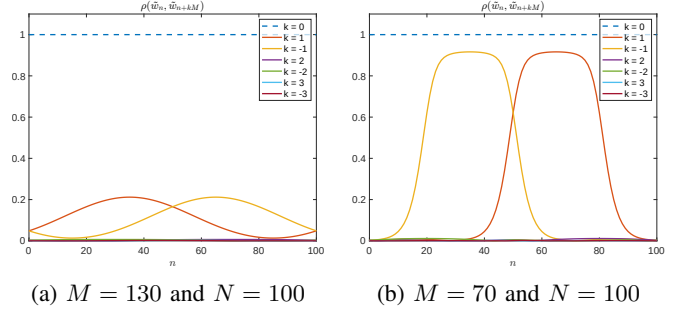


Fig. 6: $\rho(\tilde{w}_n, \tilde{w}_{n+kM})$ for different small values of k in Eq. 14 in the case of downsampling (Fig. 6a) and upsampling (Fig. 6b). $\rho(\tilde{w}_n, \tilde{w}_{n+kM})$ is non-negligible only for $k = 0, \pm 1$ and vanishes rapidly as $|k|$ increases. The length of the segments is 11.

different small values of k in the case of downsampled and upsampled white noise. Similarly to Fig. 5, we can see the correlation between two segments vanishes as $|k|$ increases and is the strongest at $|k| = 1$.

The same remark is valid in 2D. Consider a 2D signal of size $N_1 \times N_2$ resampled from an original 2D white noise of size $M_1 \times M_2$, and denote the DFT of the resampled signal by $\tilde{v}$. Let $k_1, k_2 \in \{\pm 1\}$. Let $\mathcal{I}_l = [-l, l]$ be a small interval. Then the means of two small patches separated by $(k_1 M_1, k_2 M_2)$ components,

$$\tilde{w}_{n_1, n_2} = \frac{1}{l^2} \sum_{\Delta_1 \in \mathcal{I}_l} \sum_{\Delta_2 \in \mathcal{I}_l} \tilde{v}_{n_1+\Delta_1, n_2+\Delta_2}$$

and

$$\tilde{w}_{n_1+k_1 M_1, n_2+k_2 M_2} = \frac{1}{l^2} \sum_{\Delta_1 \in \mathcal{I}_l} \sum_{\Delta_2 \in \mathcal{I}_l} \tilde{v}_{n_1+k_1 M_1+\Delta_1, n_2+k_2 M_2+\Delta_2}$$

are highly correlated for most (n_1, n_2) , which can be measured by calculating the PCC of the two patches.

We illustrate the correlations in the spectrum of a resampled image in Fig. 1, where the image has been resampled from 512×512 to 665×665 . After preprocessing, two patches in the spectrum separated by $153 = -512 \pmod{665}$ components along the height have similar patterns, showing their strong correlation in between.

III. A *contrario* DETECTION OF IMAGE RESAMPLING

In this section, we propose an *a contrario* method to detect image resampling by analyzing the anomalous correlations within the spectrum caused by resampling, as pointed out in Sec. II. Given an image under scrutiny, we first preprocess it with a non-linear filter, which aims at removing all natural structures and keeping only a residual, see Fig. 7. As we will see, the residual signal usually has statistics similar to white noise. Then, we detect anomalous correlations between patches of the residual spectrum.

A. A *contrario* detection of anomalous local correlations

Suppose that an image of size $M_1 \times M_2$ is resampled to an image of size $N_1 \times N_2$. In Sec. II-B, we have shown that

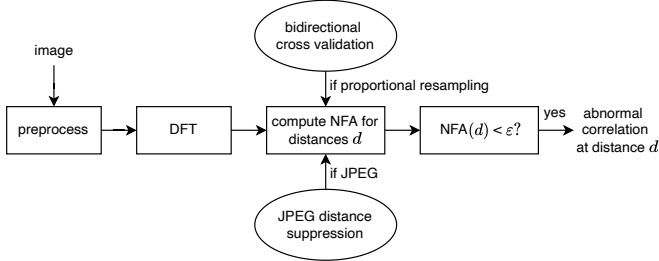


Fig. 7: The pipeline of the *a contrario* detection of image resampling by detecting anomalous correlations in the spectrum at distance d .

small patches of the spectrum separated by $k_1 M_1$ and $k_2 M_2$ frequency components along the first and second axis are correlated, where k_1 and k_2 are two arbitrary integers, and the correlation vanishes as $|k_1|$ or $|k_2|$ increases and becomes too weak to be detected. In the following, we shall describe the resampling detection along one axis, being the application to the other axis analogous. Taking into consideration the period (N_1, N_2) of the spectrum in scrutiny, we therefore aim at detecting correlated patches that are separated by $\pm M_1 \pmod{N_1}$ or $\pm M_2 \pmod{N_2}$ frequency components along the first or second axis, respectively.

A detection threshold is necessary to point out the anomalous local correlations in the spectrum of a resampled image. Such thresholds will be obtained by the non-parametric *a contrario* framework. The *a contrario* framework is based on a general perception principle, the Helmholtz principle [14], which states that a structure is perceived whenever some large deviation from randomness occurs. The structure to perceive is a pair of correlated spectrum patches separated by $d = M_1 \pmod{N_1}$ or $d = -M_1 \pmod{N_1}$ frequency components along the axis.

The proposed *a contrario* method is based on the null hypothesis that the image has not been resampled. In the residual of such an image, given an arbitrary distance d , there is no particular reason to observe a large number of correlated patch components. Thus, if we observe a large number of such correlated pairs of spectrum patches, the image is very likely to have been resampled. The linear correlation between spectrum patches is measured by the Pearson Correlation Coefficient (PCC). Since the spectrum patches are complex-valued, the PCC should also be defined accordingly. Let $\mathbf{x}, \mathbf{y} \in \mathbb{C}^N$ be two vectorized local patches, the Pearson correlation coefficient between $\mathbf{x}$ and $\mathbf{y}$ is defined by

$$\text{corr}(\mathbf{x}, \mathbf{y}) := \frac{|\langle \mathbf{x} - m(\mathbf{x}), \mathbf{y} - m(\mathbf{y}) \rangle|}{\|\mathbf{x} - m(\mathbf{x})\|_2 \|\mathbf{y} - m(\mathbf{y})\|_2}, \quad (15)$$

where $m(\cdot)$ denotes the mean of a vector, $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=0}^{N-1} x_i \bar{y}_i$ the inner product between $\mathbf{x}$ and $\mathbf{y}$, and $\|\cdot\|_2$ the ℓ_2 -norm of a vector.

We must now decide whether a measured PCC correlation between two patches is significantly large to be anomalous. This can be decided from the image spectrum itself by a non-parametric comparison between the measured correlation and

the observed background correlations from other random pairs of patches.

Let us denote by $\tilde{v}$ the DFT of the tested image. Let $I_i = \llbracket n_1^i, n_1^i + l_1 - 1 \rrbracket \times \llbracket n_2^i, n_2^i + l_2 - 1 \rrbracket$ be a small patch of frequency entries, with $l_1 \ll N_1$ and $l_2 \ll N_2$. Let $P_i = (\tilde{v}_{n_1, n_2})_{(n_1, n_2) \in I_i}$ be an anchor patch of the spectrum, and $Q_i = (\tilde{v}_{n_1+d, n_2})_{(n_1, n_2) \in I_i}$ be the patch that is d components apart from P_i in the first axis. Denote by $\rho(P_i, d) := \text{corr}(P_i, Q_i)$ the correlation of the anchor patch P w.r.t. another patch lying d coefficients apart. Under the null hypothesis H_0 , the correlations for an interval of $2r + 1$ distances, $\mathcal{N}_r(P_i, d) := \{\rho(P_i, d') : d' \in \llbracket d-r, d+r \rrbracket\}$, should have similar values. The alternative hypothesis is that the image has been resampled and that for the particular distance d , $\rho(P_i, d)$ is very likely to be a local maximum in $\mathcal{N}_r(P_i, d)$.

This can be tested by counting the number of maximum correlations among all patch pairs. To this end, we divide the domain of $\tilde{v}$ into small non-overlapping patches $\mathcal{P} = \{P_i\}$. The number of correlation maxima among all considered patch pairs is

$$k(d) := \sum_{P_i \in \mathcal{P}} \mathbb{1}_{\{\rho(P_i, d) = \max \mathcal{N}_r(P_i, d)\}}. \quad (16)$$

If $k(d)$ is larger than a certain anomaly threshold κ , then we shall say that there is an unusual large number of correlated patch pairs at the distance d in the spectrum.

Instead of directly fixing the anomaly threshold κ , the *a contrario* framework controls the number of falsely detected distances d just by chance in a random spectrum. This depends on the number of tested distances. Since the spectrum is N_1 -periodic along the axis of detection, it is enough to test all distances d satisfying $d < N_1$. In practice, we shall ignore the distance candidates that are too close to the spectrum end points, as they do not have enough neighboring distances to validate the maximum correlation. Therefore, we only test distance candidates in the restricted domain $\mathcal{C} = \llbracket r+1, N_1-r-1 \rrbracket$.

Denote the random variable $k(d)$ by $\mathbf{k}(d)$. The *a contrario* framework implicitly sets κ such that the expected number of detections is smaller than a certain ϵ under H_0 :

$$\kappa \in \{\eta : \mathbb{E}_{H_0} \left[\sum_{d \in \mathcal{C}} \mathbb{1}_{\{\mathbf{k}(d) \geq \eta\}} \right] = \#\mathcal{C} \cdot \mathbb{P}_{H_0}(\{\mathbf{k}(d) \geq \eta\}) < \epsilon\} \quad (17)$$

where $\#\mathcal{C} = N_1 - 2r - 1$ is the total number of tested distances. Taking the lower bound of the set of possible thresholds, κ is set to satisfy $\#\mathcal{C} \cdot \mathbb{P}_{H_0}(\mathbf{k}(d) \geq \kappa) = \epsilon$. We define the Number of False Alarms (NFA) of a tested distance d by

$$\text{NFA}(d) := \#\mathcal{C} \cdot \mathbb{P}_{H_0}(\mathbf{k}(d) > k(d)). \quad (18)$$

Then, verifying the threshold condition $k(d) > \kappa$ is equivalent to ensuring $\text{NFA}(d) < \epsilon$, which validates a detection with a false alarm rate smaller than ϵ .

It remains to calculate $\mathbb{P}_{H_0}(\mathbf{k}(d))$. Under H_0 , the probability that a correlation $\rho(P_i, d)$ is the maximum of $\mathcal{N}_r(P_i, d)$ is $\frac{1}{2r+1}$. Since the patches in $\mathcal{P}$ do not overlap, this ensures the independence of $\rho(P_i, d)$ for any two different P_i . Thus, under H_0 , $\mathbf{k}(d)$ follows a binomial distribution

$$H_0 : \mathbf{k}(d) \sim B\left(\#\mathcal{P}, \frac{1}{2r+1}\right) \quad (19)$$

where $\#\mathcal{P}$ is the number of non-overlapped patches. Finally, a distance d is validated if

$$\text{NFA}(d) = (N_1 - 2r - 1) \cdot \mathbf{B}\left(k(d), \#\mathcal{P}, \frac{1}{2r+1}\right) < \epsilon \quad (20)$$

where $\mathbf{B}$ is the tail of the binomial law, $\mathbf{B}(k, n, p) = \sum_{i=k}^n \binom{n}{i} p^i (1-p)^{n-i}$. The NFA is computed for all distances $d \in \mathcal{C}$, and the image is classified as a resampled image if at least one distance d is validated.

It is worth noting that once an anomalous d is detected, the original size M_1 cannot be determined due to the ambiguity of M_1 in $d = \pm M_1 (\bmod N_1)$, but d reduces the possible values of M_1 to a limited set $\{\pm d + kN_1, k \in \mathbb{Z}\} \cap \mathbb{N}_+^*$, which is an important clue of the original size.

B. Preprocessing the image to avoid false detections

In Sec. II we showed that, for white noise, spectral correlations are present when resampling. However, natural images differ largely from white noise; therefore, the conclusion of spectral correlations does not necessarily hold. In addition, the scene content in natural images can also introduce correlations between patch pairs and cause false positive detections. This usually happens on images containing replicated patterns. A simulated example with two squares of the same size and additive Gaussian noise is shown in Fig. 8. Replicated patterns are present in the spectrum and cause strong local correlations at specific distances. Fig. 9 shows in blue a series of false positive distances detected with significantly low NFAs.

To suppress false detections caused by the image contents, we apply a preprocessing filter prior to detecting image resampling. The preprocessing filter must be able to suppress most of the scene contents while preserving the subtle resampling traces in the residuals. Moreover, the preprocessing filter should be able to homogenize the spectrum such that it resembles a spectrum of resampled white noise to make the conclusion of spectral correlations hold again. Two preprocessing filters seem particularly suitable to homogenize the spectrum and therefore suppress false detections:

- **Rank transform** [53] The rank transform R of an image I computes the rank of the intensity of each pixel x in its 7×7 neighborhood $\mathcal{N}(x)$:

$$R(x) = \sum_{x' \in \mathcal{N}(x)} \mathbb{1}_{\{I(x') < I(x)\}} \quad (21)$$

- **Total variation (TV) denoiser** [10][15] The TV denoiser extracts the bounded variation component of each image, which is piecewise smooth with smooth discontinuities. The image residual contains the image noise and the tiny oscillations caused by resampling.

The effects of the above two preprocessing filters on the detections on the previous simulated image are shown in Fig. 9. We note that the rank transform and the TV denoiser succeed in suppressing the false detection present in the image without preprocessing. We found that the rank transform is more suitable for suppressing false detections on uncompressed images, while the noise extraction by the TV denoiser [15] is more suitable on JPEG-compressed images. A comprehensive

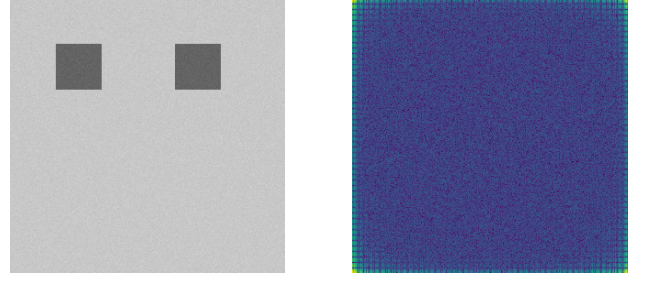


Fig. 8: Left: a simulated image with two squares of the same size and additive Gaussian noise. Right: the spectrum of the image. Periodic patterns are present in the spectrum due to the replicated squares and can lead to anomalous local correlations between frequency components at a certain fixed distance.

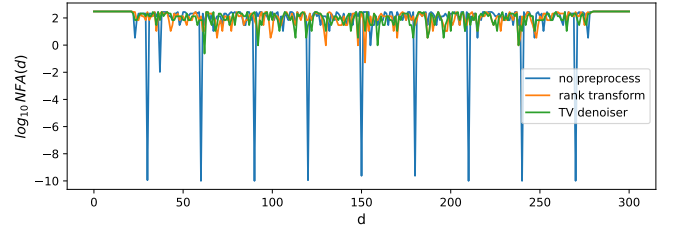


Fig. 9: NFAs of all the tested distances on the simulated image in Fig. 8, after preprocessing the image by the rank transform, extracting its TV noise, or without preprocessing. The image has undergone no resampling, but still some distances with significantly low NFAs are detected when no preprocessing is applied. These false detections are no longer present in the residual of the TV denoiser or in the image processed by the rank transform.

comparison between different preprocessing filters will be presented in Sec. IV-D.

C. Suppression of JPEG traces

It has been shown in previous works [33][54][19] that the JPEG compression using 8×8 blocks leaves traces that are similar to a resampling by factor 8. This can be understood by considering the extreme case where JPEG compression only keeps the DC component (the mean) of each block. Hence the compressed image is the result of a) zooming down the image by an 8 factor using the 8×8 block filter and b) zooming in the image by 8 using the nearest neighbor interpolation filter. It is easily checked that these operations entail high spectral correlations peaks at distances $d \in \{\frac{kN_1}{8}, k = 1, \dots, 7\}$. Besides, one can observe that in JPEG images the correlations at distances close to $\{\frac{kN_1}{8}, k = 1, \dots, 7\}$ are also slightly higher than other correlations. These high spectral correlations result in false detections, and also hinder the validation of any distance related to image resampling whose comparison interval contains the high JPEG correlations.

In order to suppress the distances related to JPEG compression, we exclude the distances $d \in \{\frac{kN_1}{8}, k = 1, \dots, 7\}$ from the tested distance candidates $\mathcal{C}$ and also from the intervals of distances used for validating maximum correlations when

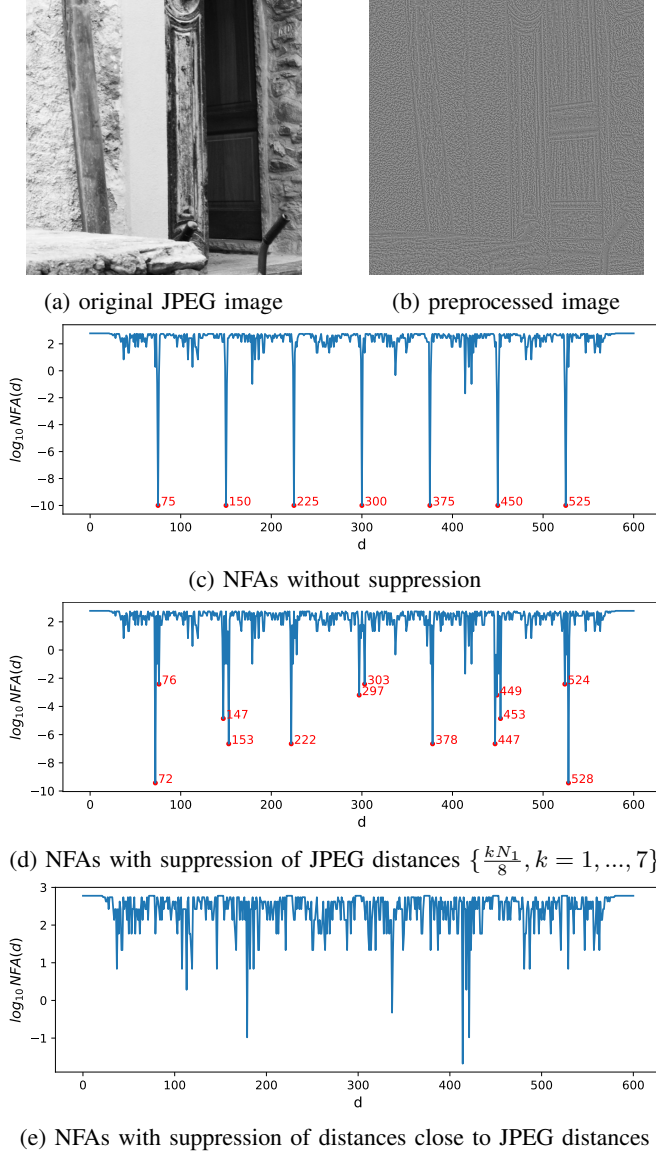


Fig. 10: A JPEG-compressed image (Fig. 10a) without resampling is preprocessed with TV denoiser (Fig. 10a). The NFAs of $\{\frac{kN_1}{8}, k = 1, \dots, 7\}$ are significantly low (Fig. 10c) if no suppression of JPEG distances is applied. If we only exclude the JPEG distances when computing NFAs, some distances close to $\{\frac{kN_1}{8}, k = 1, \dots, 7\}$ are still significant (Fig. 10d), which are removed by additional suppression (Fig. 10e).

computing the NFAs. Then among the distances whose NFAs are lower than the threshold ϵ , we empirically suppress the distances $\bigcup_{k=1}^7 [\frac{kN_1}{8} - 3, \frac{kN_1}{8} + 3]$ that are no further than 3 components from the previously suppressed distances. An example of the suppression of JPEG distances in a JPEG image without resampling is shown in Fig. 10.

D. Cross validation for detecting proportional resampling

Image rescaling mostly occurs in equal proportions along the width and height, to keep the same image aspect ratio. If we assume such proportional resampling, then spectral correlation distances are expected to be proportional in both directions.

This allows us to further eliminate false detections due to the image contents. Indeed, the false positive distances caused by image contents are usually independent in both directions.

Consider an image of original size $M_1 \times M_2$, resampled at the same resampling rate q in both directions to size $N_1 \times N_2$, with $N_1 = qM_1$ and $N_2 = qM_2$. The expected positive distances along the first axis are $\mathcal{D}_1 = \{kM_1 \bmod qM_1 = N_1(\frac{k}{q} - \lfloor \frac{k}{q} \rfloor), k \in \mathbb{Z}\}$ and along the second axis are $\mathcal{D}_2 = \{N_2(\frac{k}{q} - \lfloor \frac{k}{q} \rfloor), k \in \mathbb{Z}\}$, which have the same ratios $\{\frac{k}{q} - \lfloor \frac{k}{q} \rfloor, k \in \mathbb{Z}\}$ to N_1 and N_2 , respectively. In practice, after the NFAs are computed along both axes to obtain $\{\text{NFA}_1(d_1), d_1 < N_1\}$ and $\{\text{NFA}_2(d_2), d_2 < N_2\}$, a distance along one axis is validated as positive if the distance along the other axis related to the same ratio is also positive. The cross validation along one axis is processed by

$$\text{NFA}_1^{\text{valid}}(d_1) = \max \left(\text{NFA}_1(d_1), \text{NFA}_2 \left(\frac{d_1 N_2}{N_1} \right) \right) \quad (22)$$

We relax the precondition of proportional resampling by allowing a small difference $\beta \ll 1$ between the ratios in the two directions. Then the NFA after cross validation on the first axis is

$$\text{NFA}_1^{\text{valid}}(d_1) = \max \left(\text{NFA}_1(d_1), \min_{\left| \frac{x}{N_2} - \frac{d_1}{N_1} \right| < \beta} \text{NFA}_2(x) \right). \quad (23)$$

IV. EXPERIMENTS

Experiments were carried out using 1000 uncompressed images from the Raise-1k dataset [13]. As the demosaicing traces are similar to oversampling traces by a factor 2, the images were first downsampled by a factor 2 to remove these traces. Then, the images were resampled at different factors and cropped to 600×600 in size, resulting in 1000 resampled images for each resampling factor. Meanwhile, the same demosaiced images without resampling were directly cropped to 600×600 to serve as original images without resampling. Experiments were performed on greyscale images, the extension to color being straightforward.

A. Aliased images

We first compared our method with five unsupervised resampling detection methods in the literature: Popescu [42], Gallagher [19], Birajdar [7], Mahdian [35], and Qiao [44]. All of these methods require a threshold to distinguish original and resampled images. We set a detection threshold ensuring a false positive rate of 1% for each method, and compared their detection accuracies. All these compared methods including ours are unsupervised.

The comparison was carried out on images resampled at factors between 0.6 and 1.6, as downsampled images at a factor below 0.5 are almost undetectable and upsampled images at a factor above 1.6 are easy to detect. No anti-aliasing was applied for downsampling. The resampled images were saved respectively in uncompressed PNG format and JPEG format at qualities 95 and 90. For the PNG images, our method uses the rank transform as preprocessing. The JPEG images are instead preprocessed by a TV denoiser, and the detection

	Resampling factor	0.6	0.7	0.8	0.9	1.1	1.2	1.3	1.4	1.5	1.6	average
PNG	Qiao [44]	0.3	0.5	0.6	0.9	2.3	3.5	5.7	9.5	17.0	25.1	6.5
	Mahdian [35]	4.5	25.1	3.5	12.5	62.5	90.8	98.2	98.1	93.3	96.7	58.5
	Gallagher [19]	4.7	25.4	34.7	37.0	68.0	93.1	98.5	98.4	96.0	98.1	65.4
	Birajdar [7]	2.1	31.0	16.3	46.0	95.9	98.4	99.2	99.4	98.6	99.3	68.6
	Popescu [42]	30.0	77.3	73.9	1.3	16.8	97.5	98.9	99.1	98.7	98.8	69.2
	Ours	<u>93.1</u>	<u>99.9</u>	<u>100.0</u>	<u>99.8</u>	<u>99.7</u>	<u>99.9</u>	<u>99.8</u>	<u>99.8</u>	<u>99.8</u>	<u>99.8</u>	<u>99.2</u>
	Ours, cross-val.	98.1	100.0	99.9	99.9	100.0	99.9	100.0	99.9	100.0	100.0	99.8
JPEG Q=95	Qiao [44]	0.3	0.8	-	0.8	1.6	1.9	2.3	3.6	4.0	-	1.9
	Mahdian [35]	3.6	20.1	-	9.7	51.2	77.1	93.0	93.2	83.6	-	53.9
	Gallagher [19]	3.9	25.5	-	33.4	66.7	87.6	96.1	95.6	90.5	-	62.4
	Birajdar [7]	1.4	11.6	-	15.5	83.9	87.3	93.5	97.0	93.0	-	60.4
	Popescu [42]	0.0	0.6	-	1.1	1.3	2.6	1.3	2.4	5.7	-	1.9
	Ours	<u>48.0</u>	<u>78.1</u>	-	<u>93.0</u>	<u>98.0</u>	<u>98.9</u>	<u>97.5</u>	<u>98.1</u>	<u>99.6</u>	-	<u>88.9</u>
	Ours, cross-val.	60.0	88.4	-	93.0	97.5	99.1	97.5	98.6	99.6	-	91.7
JPEG Q=90	Qiao [44]	0.4	0.8	-	0.8	1.4	1.7	1.8	2.6	3.2	-	1.6
	Mahdian [35]	2.8	13.8	-	5.8	38.4	58.9	77.5	83.8	73.3	-	44.3
	Gallagher [19]	3.3	13.8	-	15.3	44.0	63.9	77.6	84.7	77.6	-	47.5
	Birajdar [7]	2.2	6.6	-	9.6	58.3	65.3	77.2	87.4	75.1	-	47.7
	Popescu [42]	0.0	0.0	-	0.3	0.7	0.1	0.0	0.3	0.2	-	0.2
	Ours	<u>26.6</u>	<u>41.3</u>	-	<u>66.8</u>	<u>88.2</u>	<u>93.1</u>	<u>83.9</u>	<u>86.9</u>	<u>96.5</u>	-	<u>72.9</u>
	Ours, cross-val.	36.0	55.8	-	66.1	89.0	93.0	88.6	90.6	96.1	-	76.9

TABLE III: Detecting accuracy (%) of different detection methods at false positive rate of 1% and at different resampling factors from 0.6 to 1.6. Resampled images are saved in PNG format and JPEG format at qualities 95 and 90. The image resampling at factors 0.8 and 1.6 are ignored because the correlation peaks due to resampling overlap with those caused by JPEG compression.

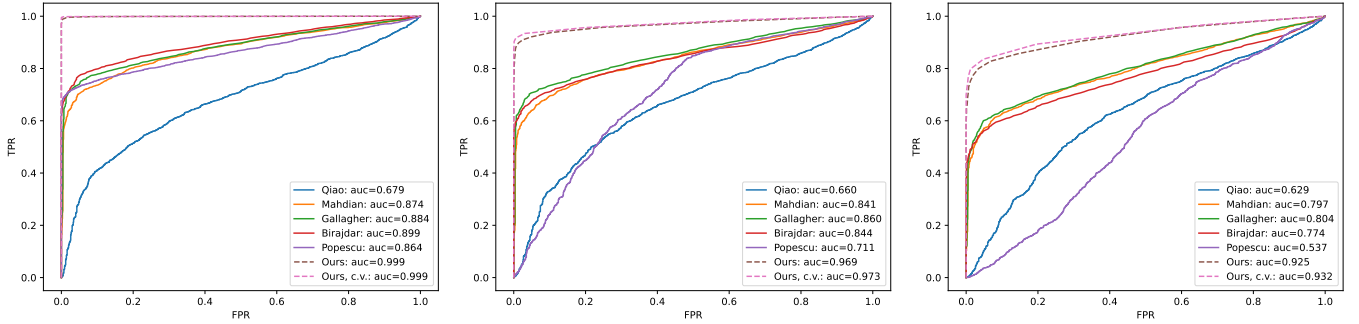


Fig. 11: ROC curves and AUCs of different resampling detection methods on images resampled at resampling rates from 0.6 to 1.6. Left: the images are saved in PNG; middle: the images are compressed by JPEG at quality factor 95; right: the images are compressed by JPEG at quality factor 90. Our method is highlighted in dashed curves.

process is applied to the extracted noise. The performance of our method was evaluated on resampling detection in the horizontal direction, and also on resampling detection with cross validation, assuming that the resampling ratio is the same in both horizontal and vertical directions.

The results are shown in Tab. III. Our method outperforms all the compared methods for all the resampling factors and compression qualities. For uncompressed PNG images, our method without cross-validation attains an accuracy over 93% at resampling factor 0.6 and detects almost all the resampled images at higher resampling factors, substantially ahead of all the compared methods. Using cross-validation to suppress false positives, the detection accuracy of the resampling factor 0.6 is increased to 98.1%. When the resampled images are saved in JPEG at quality factor 95, the detection accuracies are still higher than 97% for upsampled images, but drop significantly on downsampled images compared to uncompressed images. The cross-validation is helpful to boost the detection

performance of downsampled images. At JPEG quality factor 90, the cross-validation only has limited effect on accuracy, because the resampling traces are severely altered by JPEG compression. Nevertheless, the performance of our method is still superior to the compared methods on strongly compressed images.

The comparative receiver operating curves (ROC) and the areas under the curve (AUC) for uncompressed images and JPEG-compressed images at quality factors 95 and 90 are shown in Fig. 11. Each curve counts the original images as true samples and all resampled images at all resampling factors between 0.6 and 1.6 as false samples. We can see that our method outperforms the others by a large margin, having high recalls of about 100%, 90% and 80%, respectively for the three compression configurations at very low false positive rates. This is attributed to the proposed *a contrario* detection method of anomalous local spectral correlations, that only reveals significant resampling traces, and to the preprocessing

Resampling factor	0.6	0.7	0.8	0.9	1.1	1.2	1.3	1.4	1.5	1.6	average	$[0.6, 0.95] \cup [1.05, 1.6]$
ConstNet-q	0.2	0.3	1.2	2.9	26.2	49.3	77.1	95.4	97.5	98.0	44.8	42.2
ConstNet-r	0.0	0.1	0.4	3.5	21.5	41.1	62.5	86.0	94.7	96.1	40.6	39.9
Ours	17.9	32.2	68.6	79.1	98.1	98.5	99.2	99.5	99.4	99.6	79.2	74.6

TABLE IV: Detection accuracy (%) of our method and ConstNet [6] at false positive rate of 1% on 256×256 images having been resampled at different factors from 0.6 to 1.6. Two versions of ConstNet trained with quantized resampling factors in $\{0.6, 0.7, \dots, 1.6\}$ (ConstNet-q) and with random real resampling factors in $[0.6, 0.95] \cup [1.05, 1.6]$ (ConstNet-r) are compared. The middle columns show the accuracies on test images resampled at quantized factors, and the last column shows the accuracies for real resampling factors. Bicubic interpolation is being applied, and no anti-aliasing is processed. Test images are saved in PNG.

step that efficiently suppresses false positives.

In addition, we compared our method with a supervised method based on convolutional network, ConstNet [6], on images of fixed size 256×256 which is the input size required by ConstNet. Considering that ConstNet deals with resampling detection as a classification problem which relies on the pre-defined classes, two versions of ConstNet were compared: ConstNet trained on images resampled at quantized factors in $\{0.6, 0.7, \dots, 1.6\}$ (ConstNet-q), and ConstNet trained on images resampled at random real factors in $[0.6, 0.95] \cup [1.05, 1.6]$ (ConstNet-r). The resampling factors too close to 1 were neglected as the related resampling traces are almost undetectable. 5K original and 5K resampled images were created for training using images from the IEEE IFS-TC image forensics challenge, as performed in [6]. Bicubic interpolation was used for resampling and no post-JPEG compression was applied. Both versions of ConstNet and our method were then evaluated on images from the Raise-1k dataset [13] resampled at the same quantized factors and real factors. The evaluation of ConstNet on a dataset different from the one used for training essentially eliminated the potential advantage from the data bias in the same dataset to guarantee a fair comparison, since our method does not rely on any training data and is naturally not affected by the data bias.

The results are shown in Tab. IV, where we can see our method outperforms ConstNet at all the resampling factors. Both of them have poor performance on detecting downsampled images and have detecting accuracies higher than 90% only when the resampling factor is higher than 1.4, implying that ConstNet prioritizes the detection of upsampled images and generalizes poorly on all the resampling rates. An explanation is that a convolutional network like ConstNet composed of a limited number of convolutional kernels only focuses on and prioritizes a certain subset of frequency bands and is unable to perform equally well on all the frequency bands. According to the analysis in Sec. II, image resampling results in artifacts of spectrum folding that can occur at any frequency band depending on the resampling factors, which is intractable for a convolutional network. In contrast, our method invariant to frequency bands by design performs robustly for different resampling factors. Besides, our method is easily extended to any size of images, while ConstNet requires a fixed input size and is inconvenient for detecting images of a different size.

B. Anti-aliased images

In order to prevent aliasing in the downsampled image, a low-pass filter is usually applied to the image before downsampling.

If an image has undergone a strictly band-limited filtering, no frequency should be folded in the downsampled image, and the image downsampling would no longer be detectable. Applying a strictly band-limited anti-aliasing filter is equivalent to a convolution with a sinc filter in the spatial domain, which results in ringing effects. A smooth frequency cut-off therefore is usually preferred, as a compromise between ringing and aliasing. Such a strategy entails some frequency folding that can still be detected by our method. Hence, two representative anti-aliasing strategies were considered to compare the performances of different methods: 1) anti-aliasing with Gaussian blur; 2) anti-aliasing by stretching the support of interpolation kernel.

Gaussian blur Gaussian kernel is widely used as anti-aliasing filter. Assuming that an initial optical Gaussian blur with standard deviation σ_a is sufficient to ensure an approximately well-sampled image, a relative blur with $\sigma = \sigma_a \sqrt{1/r^2 - 1}$ is required to maintain well-sampling for the resampled image by a factor $r < 1$. In our experiments, we conservatively assumed $\sigma_a = 0.8$ which yields a slightly blurry image, compared to the practical value of $\sigma_a = 0.5$ assumed for the SIFT keypoint extraction [34]. This allowed us to evaluate the downsampling detection performance after a very conservative anti-aliasing. This comparison was carried out on images resampled at resampling factors between 0.6 and 0.95 with the bicubic interpolator and saved in PNG. Its results are shown in Tab. V. As was expected, our detector performs generally worse on anti-aliased images than on aliased images. Surprisingly, four of the compared methods Mahdian [35], Gallagher [19], Birajdar [7] and Popescu [42] show a better accuracy at certain resampling factors on anti-aliased images than on aliased images. Nevertheless, our proposed method still outperforms the compared methods at all downsampling factors.

Kernel stretching The second classic anti-aliasing filter was achieved by convolving the image with a interpolation kernel whose spatial support is stretched [45]. Let φ be the interpolation filter defined in the spatial domain. To downsample an image with resampling factor $r \in (0, 1)$, the interpolation filter is transformed by stretching its support by a factor $1/r$

$$\varphi_r(x) = r \cdot \varphi(rx) \quad (24)$$

before convolving it with the original image. This is equivalent to reducing the bandwidth by a factor $1/r > 1$ in the frequency domain, therefore attenuating the frequencies above the band limit. We applied our comparison on images resampled with the bicubic interpolator and saved in PNG. Its results are shown

Resampling factor r	0.6	0.65	0.7	0.75	0.8	0.85	0.9	0.95	average
Qiao [44]	1.4	1.3	1.2	0.9	1.3	1.0	1.1	1.2	1.2
Mahdian [35]	47.6	32.5	1.1	6.4	2.4	3.7	11.2	3.3	13.5
Gallagher [19]	52.4	30.9	1.2	7.0	11.7	21.4	31.8	20.4	22.1
Birajdar [7]	43.3	57.9	22.0	39.1	10.0	15.0	37.6	75.5	37.6
Popescu [42]	60.6	12.5	1.3	2.1	0.8	0.5	1.1	0.8	10.0
Ours	88.0	91.1	81.1	97.0	95.8	95.7	99.7	99.5	93.5
Ours, cross-val.	96.5	97.9	94.5	99.5	98.9	97.4	99.8	99.4	98.0

TABLE V: Detecting accuracy (%) of different detection methods at false positive rate of 1% on images downsampled with anti-aliasing by Gaussian blur ($\sigma = 0.8\sqrt{1/r^2 - 1}$) at different factors r from 0.6 to 0.95, with bicubic interpolation.

Resampling factor	0.6	0.65	0.7	0.75	0.8	0.85	0.9	0.95	average
Qiao [44]	1.4	1.3	1.2	0.9	1.3	1.0	1.1	1.2	1.2
Mahdian [35]	1.7	22.2	6.1	18.1	1.9	7.6	9.9	3.6	8.9
Gallagher [19]	2.1	21.6	6.3	21.8	15.2	20.2	25.4	20.2	16.6
Birajdar [7]	3.1	3.6	5.7	12.4	3.5	23.0	56.8	85.0	24.1
Popescu [42]	5.5	76.1	21.7	49.6	29.5	25.8	6.0	2.4	27.1
Ours	31.0	35.5	35.6	41.1	66.8	86.4	96.4	97.5	61.3
Ours, c.v.	24.5	29.9	29.3	41.0	73.3	89.1	96.7	98.1	60.2

TABLE VI: Detecting accuracy (%) of different detection methods at false positive rate of 1% on images downsampled with anti-aliasing by spatial-domain kernel stretching (see Eq. 24) at factors from 0.6 to 0.95, with bicubic interpolation.

in Tab. VI. As can be seen, the resampling detection is much harder. All methods perform globally worse than on images filtered by Gaussian blur. Interestingly though, Popescu and Farid’s [42] method performs better than our method for the specific resampling factors 0.65 and 0.75, but worse for all other resampling factors. One possible explanation is that the resampling factors $0.65 \approx 2/3$ and $0.75 = 3/4$ result in a third or a fourth of non-interpolated pixels along each axis in the resampled images, and that the Popescu and Farid’s method, involving a classification between interpolated and non-interpolated pixels, is advantageous in detecting resampled images containing a large amount of non-interpolated pixels. When most pixels are interpolated for the other resampling factors, the performance of this method drops significantly while our method still works. Our method works well at higher resampling factors and globally outperforms the compared methods on resampled images with anti-aliasing by kernel stretching.

C. Robustness to different interpolating filters

In this section, we evaluate the proposed *a contrario* detection method on images resampled with different interpolators, including the nearest-neighbor, bilinear, bicubic and Lanczos interpolators. The downsampled images were aliased images. All the tested images were saved in PNG. With a threshold set for FPR at 1%, the detection accuracies at different resampling factors for each interpolator are shown in Fig. 12. We can see that image resampling with the bilinear interpolator is detectable with an accuracy of almost 100%; the performance for the bicubic interpolator is very similar. The nearest-neighbor resampling is easily detected, except for downsampling at factors close to 1. The closer the factor to 1, the more difficult it is to detect. This is because at a resampling factor close to 1 the nearest-neighbor resampling only affects very few pixels by replacing them with their adjacent pixels, and most

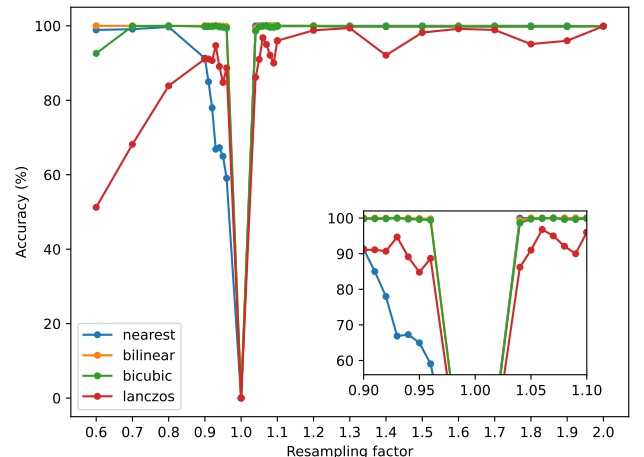


Fig. 12: Detecting accuracies on images resampled with different interpolators at different resampling factors, with an 1% FPR threshold.

of the pixels remain unchanged. As for the Lanczos interpolator, which is designed to approximate the sinc filter and attenuates a large amount of aliased frequencies, image resampling is still detectable with lower accuracies. Its detectability increases when the resampling factor increases, so that more than 90% of the upsampled images with the Lanczos interpolator can be detected.

D. Ablation study of preprocessing filters

An ablation study on the influence of different preprocessing filters on the detection performance was carried out. Five different preprocessings were compared: no preprocessing, the phase-only transform (PHOT) [1], a DCT denoiser [52], the total variation (TV) denoiser [15], and the rank transform [53]. The comparison was conducted on images resampled with the

Resampling factor		0.6	0.7	0.8	0.9	1.1	1.2	1.3	1.4	1.5	1.6	average
Uncompressed	none	14.5	50.3	92.0	85.9	98.4	99.9	99.6	98.5	99.3	99.6	83.8
	phase only transform [1]	15.6	48.0	93.6	92.2	99.2	100.0	99.9	99.1	99.4	99.7	84.7
	DCT denoiser [52]	53.9	83.2	97.2	98.1	99.5	100.0	99.9	99.9	99.8	99.8	93.1
	TV denoiser [15]	62.6	92.9	98.3	98.7	99.4	99.9	99.4	99.2	99.3	99.7	94.9
	rank transform [53]	93.1	99.9	100.0	99.8	99.7	99.9	99.8	99.8	99.8	99.8	99.2
JPEG Q=95	none	12.4	41.1	-	74.4	93.2	97.3	90.1	89.7	95.8	-	74.2
	phase only transform [1]	33.8	71.3	-	91.2	95.6	98.0	95.4	93.6	97.7	-	84.6
	DCT denoiser [52]	13.7	39.9	-	82.7	96.3	98.3	94.8	94.1	97.8	-	77.2
	TV denoiser [15]	48.0	78.1	-	93.0	98.0	98.9	97.5	98.1	99.6	-	88.9
	rank transform [53]	47.9	73.6	-	85.0	95.3	97.3	92.7	97.1	99.4	-	86.0
JPEG Q=90	none	6.3	21.5	-	42.3	73.4	86.4	67.9	64.9	79.3	-	55.2
	phase only transform [1]	8.7	24.8	-	58.9	86.1	93.1	80.6	80.8	90.8	-	65.5
	DCT denoiser [52]	21.7	45.7	-	74.9	87.2	91.0	82.0	85.2	92.3	-	72.5
	TV denoiser [15]	26.6	41.3	-	66.8	88.2	93.1	83.9	86.9	96.5	-	72.9
	rank transform [53]	30.2	46.3	-	60.5	82.0	85.3	72.5	80.1	91.6	-	68.6

TABLE VII: Detecting accuracy (%) at 1% false positive rate with different preprocessing filters and at resampling factors ranging from 0.6 to 1.6. Resampled images are respectively saved in PNG format and JPEG format at qualities 95 and 90.

bicubic interpolating filter, then saved respectively in PNG and in JPEG at quality factors 95 and 90. No cross-validation under the proportional resampling assumption was applied.

The detection accuracies with thresholding at 1% FPR are shown in Tab. VII. We can see that all the tested preprocessing filters help boost the detection performance by suppressing the undesirable image contents. Our detector combined with the rank transform performs best on uncompressed images, as the rank transform cancels out the high-level structures of an image and mainly preserves the low-level resampling traces within small neighborhoods. However, the performance on JPEG images using the rank transform is not as good as using a TV denoiser, because the rank transform, which converts a pixel value to its rank in a small neighborhood, is severely perturbed by JPEG blocking artifacts. As there are sudden changes between the pixels near the JPEG grid, the ranks of these pixels do not truly preserve the local structure and do not contain the useful resampling detection traces. On JPEG-compressed images, the detection applied on the residuals obtained by the TV denoiser achieves the best performance.

Furthermore, we studied the influence of the threshold inherent to these preprocessing filters. The threshold ϵ related to Eq. 20 defines an upper bound of the expected number of false positive distances of correlation in a random spectrum. Whenever a correlation distance is detected in a spectrum, the image is deemed resampled. Therefore, the order of magnitude of the threshold ϵ should be similar to that of the observed false positive rate if the null hypothesis is completely valid. In practice, the null hypothesis is not valid because local spectral correlations can be caused by image contents, and a more conservative value of ϵ is required to have a false positive rate of 1%. Nevertheless, as the required ϵ for false positive rate of 1% increases, the null hypothesis becomes more valid and the false positive local correlation in a normal spectrum is less significant. The thresholds set at false positive rate of 1% for these preprocessing filters and for different post-compressions are summarized in Tab. VIII. One observes that the preprocessing with the phase-only transform, TV denoising or rank transform increases the threshold value, while DCT denoising decreases the threshold value. This means that the

phase-only transform, TV denoising and the rank transform are effective in suppressing false positive local correlations caused by image contents, while DCT denoising augments the false positive local correlations. It is also seen in Tab. VII that the detection performance is actually improved when using DCT denoising, indicating that DCT denoising also augments the true positive local correlations due to image resampling. Finally, among all the tested preprocessing filters, the thresholds for the rank transform are the highest, meaning that it is the most effective filter for suppressing false positive local correlations of image contents.

Threshold ϵ at FPR=1%	PNG	JPEG Q=95	JPEG Q=90
none	1.2×10^{-15}	1.6×10^{-14}	8.5×10^{-17}
phase only transform	2.3×10^{-12}	2.6×10^{-11}	2.6×10^{-11}
DCT denoising	4.0×10^{-19}	5.8×10^{-18}	8.5×10^{-17}
TV denoising	2.2×10^{-7}	1.8×10^{-6}	1.8×10^{-6}
rank transform	1.4×10^{-5}	1.4×10^{-5}	9.7×10^{-5}

TABLE VIII: NFA thresholds set at 1% false positive rate for different preprocessing filters and different compression rates.

Qualitative analyses on several examples are shown in Fig. 13 to 16 to better illustrate the functionalities of the different preprocessing filters. The input images in Fig. 13 to 15 were resampled at factor 0.6 with the bicubic interpolator and saved in PNG. When no preprocessing, or the phase-only transform is applied, the image resampling is usually not detectable (Fig. 13 and 14). Preprocessing by noise extraction with the DCT denoiser or the TV denoiser works on images without too many textures (Fig. 13 and 15), but fail to effectively reveal the resampling traces in highly textured images (Fig. 14). Meanwhile, the rank transform is robust to different texture types and helps uncover the anomalous spectral correlation left by resampling in most cases. Fig. 16 shows the detection results on an image without resampling saved in JPEG. Normally, no anomalous distance should be detected, while false positive distances related to significantly low NFAs are detected using any of the tested preprocessing filters. This is due to the replicated pillars in the image that leave strong correlations at certain fixed distances in the spectrum. Nevertheless, the

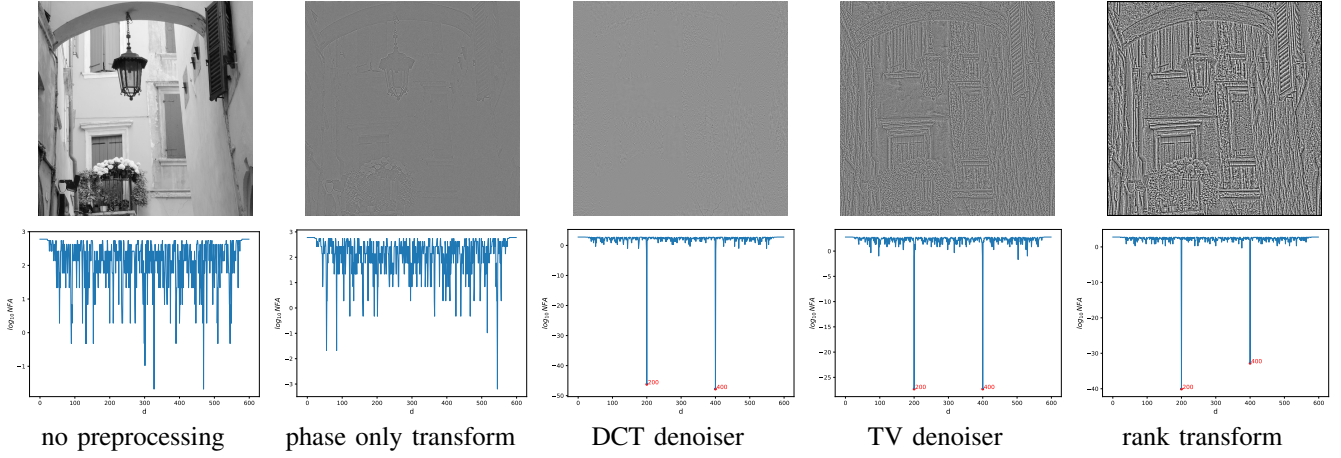


Fig. 13: Comparison with different preprocessing filters. Top: images after preprocessing; bottom: the NFAs of different distances d . The tested image is downsampled from 1000×1000 to 600×600 , and the distance d related to anomalous spectral correlations is expected at 200 and 400. The resampling is detected with DCT denoiser, TV denoiser and rank transform.

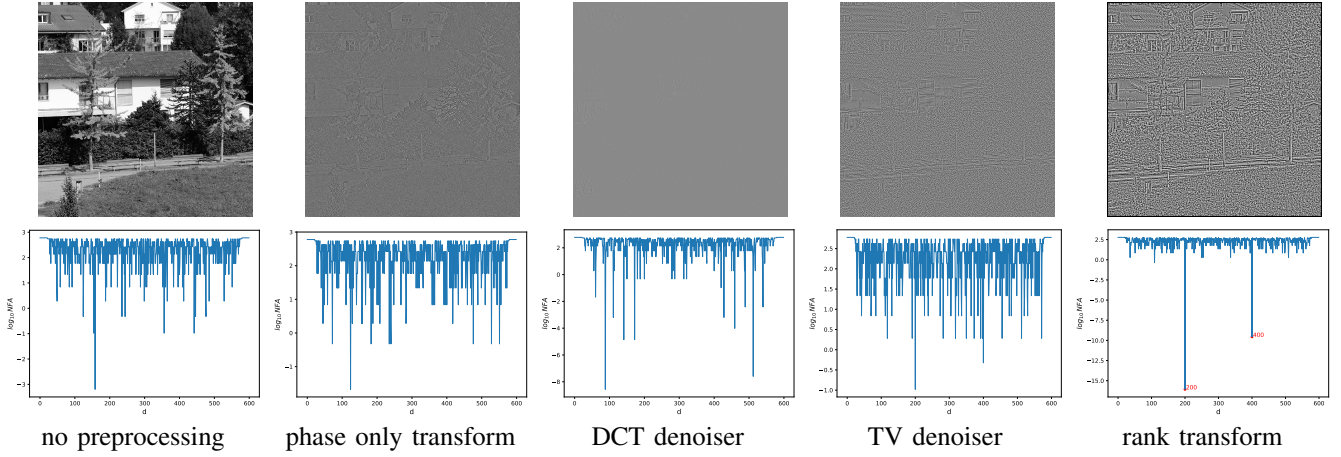


Fig. 14: Comparison with different preprocessing filters. Top: images after preprocessing; bottom: the NFAs of different distances d . The tested image is downsampled from 1000×1000 to 600×600 , and the distance d related to anomalous spectral correlations is expected at 200 and 400. The resampling is detected with the rank transform preprocessing only.

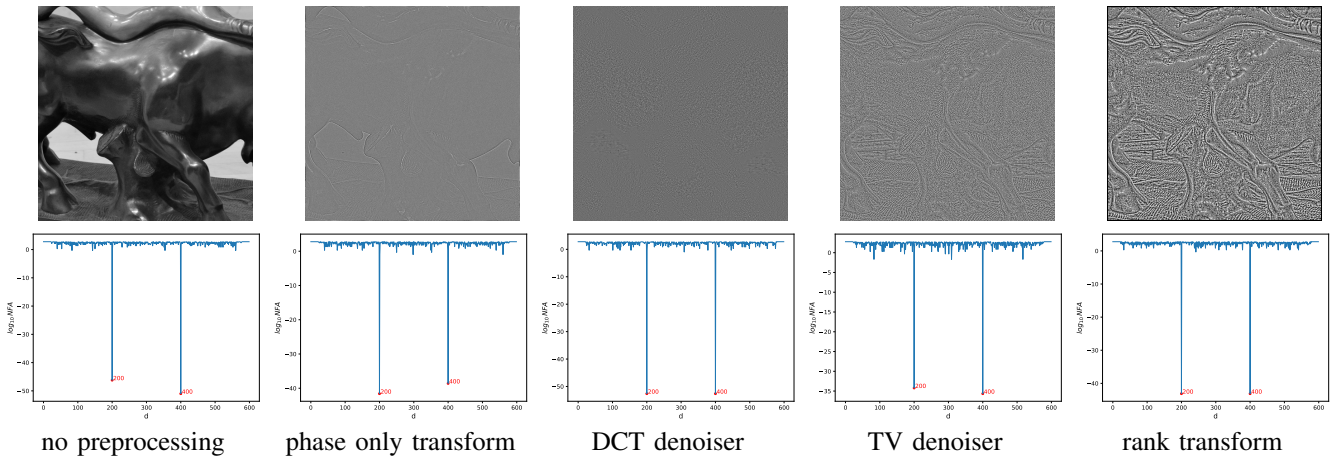


Fig. 15: Comparison with different preprocessing filters. Top: images after preprocessing; bottom: the NFAs of different distances d . The tested image is downsampled from 1000×1000 to 600×600 , and the distance d related to anomalous spectral correlations is expected at 200 and 400. The distances of correlation are detected either with or without preprocessing.

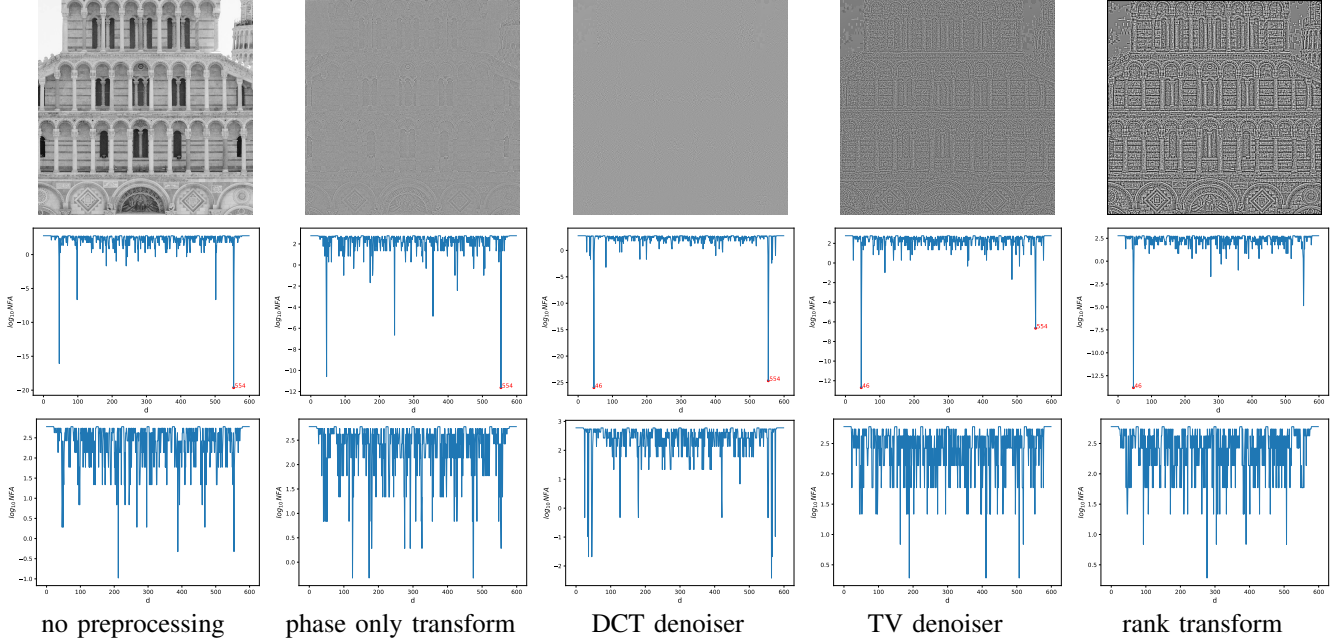


Fig. 16: Comparison with different preprocessing filters. Top: images after preprocessing; middle: the NFAs of different distances d only along the horizontal axis; bottom: the NFAs of different distances d with cross validation by assuming proportional resampling. The tested image is saved in JPEG *without* resampling. None of the preprocessing filters can suppress the false positive distances caused by the replicated pillars when the detection is processed in the horizontal direction only. With a cross validation by assuming a proportional resampling, the false positive distances of the spectral correlation are successfully suppressed.

pillars are replicated in the horizontal direction only, their resulting spectral correlations only correspond to positive distances along the horizontal but not along the vertical axis. When the cross-validation under the proportional resampling assumption is applied, these false positive distances are successfully suppressed.

E. Impact of JPEG image compression

Another evaluation was performed on resampled images with different JPEG compression levels to assess the influence of compression on the performance. The images were resampled at factors from 0.6 to 1.6 with the bicubic interpolator and without anti-aliasing, then saved in PNG or in JPEG at different quality factors Q . The detection accuracies at FPR of 1% including all the resampling factors are shown in Fig. 17. Both the results with and without cross validation are visualized. We can see that resampling is easily detected at compression quality factors above 90, and becomes difficult to detect for quality factors below 80. The improvement by cross-validation is small when the resampling detection is too easy or too difficult along each single axis, but when the resampling traces are moderately difficult for $Q=80$ or $Q=70$ the cross-validation is helpful to boost the performance.

F. Impact of image size

Finally, an evaluation was performed on resampled images of different sizes to assess the influence of image size on detection performance. The images were resampled at factors from 0.6

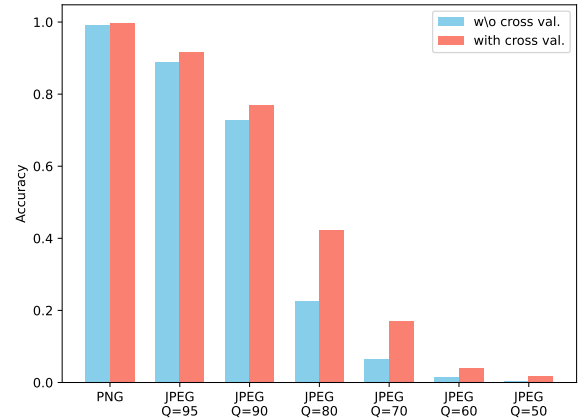


Fig. 17: Accuracy of detecting resampled images saved in PNG or compressed by JPEG at different quality factors Q . Thresholds are set for 1% false positive rate. The performances of the detection solely along the horizontal axis without cross-validation and that with cross-validation by assuming proportional resampling are both shown.

to 2.0 with the bicubic interpolator and without anti-aliasing, then saved in PNG. The detection accuracies at 1% FPR for different image sizes are shown in Fig. 18. We can see that the detection accuracy increases as the image size increases. The detection of downsampling is degraded on small images, but the detection of upsampling is achievable at an accuracy of almost 100% even on the smallest size of 300×300 .

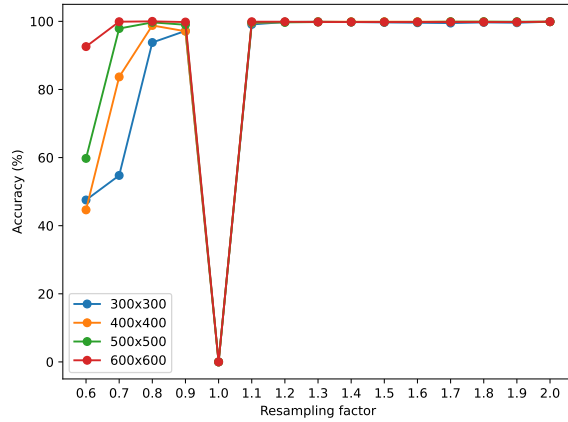


Fig. 18: Accuracy of classifying resampled images of different sizes with threshold set for 1% false positive rate for different resampling factors.

V. DISCUSSION AND CONCLUSION

In this paper, we conducted a theoretical study of the spectrum folding phenomenon for one-dimensional signal resampling and for image resampling in two dimensions. We demonstrated that the local spectral correlations caused by image resampling are exploitable for image resampling detection. Then an *a contrario* method was presented to detect anomalous correlations between the small patch pairs in the spectrum. To suppress the false positive correlation due to image contents, we proposed to utilize preprocessing filters for suppressing contents, and also cross validation for suppressing false positive distances of spectral correlation under the assumption of a proportional resampling along both horizontal and vertical directions. Experiments showed a large margin improvement of our method compared to former methods, be it for PNG images and JPEG images, and for aliased or anti-aliased images. A comprehensive study was carried out to analyze the influence of different interpolating filters used for resampling, different preprocessing filters, different JPEG compression qualities and the image size on the detection performance.

Nevertheless, the proposed method has some limitations in its current form. The presented spectrum folding argument does not allow us to determine the original size M of a resampled image from the resulting correlation offset d and the current size N due to the ambiguity of M retrieved from $d = \pm M \bmod N$. Further work is needed to find an effective disambiguation method. Also, the presented spectrum folding argument is currently limited to image rescaling where the image is resampled along each axis independently at different rates. In practice, image resampling is also used for image skewing and rotation, which is proved to be detectable in some previous works [35][42]. Our theory needs to be extended for any affine transform to address these cases.

Besides, the analysis in the frequency domain relies on the full image and is unable to detect local resampled zones. One way of detecting local resampled zones with our proposed method is to process the algorithm on small patches and

assemble the local detection results for localizing resampled zones. A last inconvenience is the high computational cost of the proposed *a contrario* algorithm, whose time complexity is $\mathcal{O}(L^3)$ for an image of size $L \times L$. This can be potentially improved by combining faster methods such as the ones in [7] [35] [19] with thresholding at high recalls to provide candidate correlation distances, then restricting the application of the *a contrario* algorithm to candidate distances.

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Longuet-Higgins Prize 2015, Doctor Honoris Causa of U. de la Republica Montevideo 2018.

APPENDIX A

THEOREM: SPECTRUM FOLDING OF 1D ALIGNED
RESAMPLING

Consider a 1D signal $\{u_k\} \in \mathbb{R}^M$ of M samples and its resampled version $\{v_k\} \in \mathbb{R}^N$ of N samples. Here $M, N \in \mathbb{N}^*$ can be any positive integers. An usual resampling method proceeds in two steps: first the samples $\{u_k\}$ are interpolated to a continuous function f using an interpolating filter φ , then f is resampled to the discrete signal $\{v_k\}$ of N samples.

$$\{u_k\} \in \mathbb{R}^M \xrightarrow[\varphi]{\text{interpolation}} f \xrightarrow{\text{resampling}} \{v_k\} \in \mathbb{R}^N$$

The interpolation from $\{u_k\}$ to a continuous function $f: \mathbb{R} \rightarrow \mathbb{R}$ is realized by convolving $\{u_k\}$ with an interpolating filter φ :

$$f(x) = \sum_{k=0}^{M-1} u_k \cdot \varphi(x-k), \quad x \in \mathbb{R}. \quad (\text{S1})$$

Then the resampling from f to $\{v_k\}$ is achieved by evenly discretizing f on $[0, M]$ to N samples:

$$v_k = f\left(k \frac{M}{N}\right), \quad k = 0, \dots, N-1. \quad (\text{S2})$$

More specifically, such resampling is an aligned resampling because the intervals for the original sequence and resampled sequence are the same. An illustrative example is shown in Fig. S1.

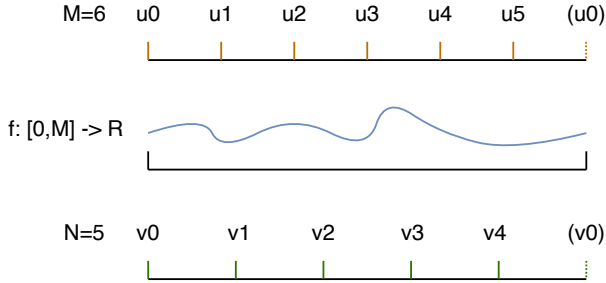


Fig. S1: Aligned resampling: the intervals for the original sequence and the resampled sequence are the same.

Theorem. Suppose the interpolating filter φ is M -periodic. Denote the DFTs of $\{u_k\}$ and $\{v_k\}$ by $\{\tilde{u}_m\} \in \mathbb{C}^M$ and $\{\tilde{v}_n\} \in \mathbb{C}^N$, respectively. Denote the Fourier coefficients of φ and f by $\{c_p(\varphi)\}_{p \in \mathbb{Z}}$ and $\{c_p(f)\}_{p \in \mathbb{Z}}$, respectively. Then we have

$$\tilde{v}_n = M \cdot \sum_{z \in \mathbb{Z}} c_{zN+n}(\varphi) \cdot \tilde{u}_{zN+n} \quad (\text{S3})$$

Proof. The interpolation process can be expanded by the DFT $\{\tilde{u}_m\}$ and the Fourier series $\{c_p(\varphi)\}$:

$$\begin{aligned} f(x) &= \sum_{k=0}^{M-1} u_k \cdot \varphi(x-k) \\ &= \sum_{k=0}^{M-1} \left(\sum_{m=0}^{M-1} \tilde{u}_m \cdot \omega_M^{km} \right) \cdot \left(\sum_{p \in \mathbb{Z}} c_p(\varphi) \cdot \omega_M^{(x-k)p} \right) \\ &= \sum_{k=0}^{M-1} \sum_{m=0}^{M-1} \sum_{p \in \mathbb{Z}} \tilde{u}_m \cdot \omega_M^{km} \cdot c_p(\varphi) \cdot \omega_M^{(x-k)p} \\ &= \sum_{p \in \mathbb{Z}} \omega_M^{xp} \underbrace{\sum_{k=0}^{M-1} \sum_{m=0}^{M-1} \tilde{u}_m \cdot \omega_M^{km} \cdot c_p(\varphi) \cdot \omega_M^{-kp}}_{c_p(f)} \end{aligned} \quad (\text{S4})$$

By identification from Eq. S4 and the Fourier series $f(x) = \sum_{p \in \mathbb{Z}} \omega_M^{xp} \cdot c_p(f)$, we have

$$\begin{aligned} c_p(f) &= \sum_{k=0}^{M-1} \sum_{m=0}^{M-1} \tilde{u}_m \cdot \omega_M^{km} \cdot c_p(\varphi) \cdot \omega_M^{-kp} \\ &= c_p(\varphi) \sum_{m=0}^{M-1} \tilde{u}_m \sum_{k=0}^{M-1} \omega_M^{k(m-p)} \\ &= M \cdot c_p(\varphi) \cdot \tilde{u}_{p \bmod M} = M \cdot c_p(\varphi) \cdot \tilde{u}_p \end{aligned} \quad (\text{S5})$$

where the last line is justified by $\sum_{k=0}^{M-1} \omega_M^{k(m-p)} = M \mathbb{1}_{\{p \bmod M=m\}}$ and $\tilde{u}_{p \bmod M} = \tilde{u}_p$.

Eq. S5 expresses that computing the Fourier coefficients of a discrete signal convolved by a continuous function amounts to multiplying the periodized DFT of the signal by the Fourier coefficients of the continuous function.

Resampling the continuous f at sampling rate $\frac{M}{N}$ to obtain the discrete sequence $\{v_k\} \in \mathbb{R}^N$ is equivalent to summing up the non-overlapped intervals in length N of the Fourier transform of f [37]: $\tilde{v}_n = \sum_{z \in \mathbb{Z}} c_{zN+n}(f)$. Combining with Eq. S5 we obtain

$$\tilde{v}_n = M \sum_{z \in \mathbb{Z}} c_{zN+n}(\varphi) \cdot \tilde{u}_{zN+n}. \quad (\text{S6})$$

□

APPENDIX B

THEOREM: SPECTRUM FOLDING OF MULTIDIMENSIONAL
ALIGNED RESAMPLING

Consider an original d -dimensional signal of size $M_1 \times \dots \times M_d$: $\mathbf{u} = \{u_{\mathbf{k}}\} \in \mathbb{R}^{M_1 \times \dots \times M_d}$ with $\mathbf{k} = (k_1, \dots, k_d)$. $\mathbf{u}$ is resampled to another d -dimensional signal $\mathbf{v} = \{v_{k_1, \dots, k_d}\} \in \mathbb{R}^{N_1 \times \dots \times N_d}$ using an interpolating filter $\varphi: \mathbb{R}^d \rightarrow \mathbb{R}^d$.

$$\{u_{\mathbf{k}}\} \in \mathbb{R}^{M_1 \times \dots \times M_d} \xrightarrow[\varphi]{\text{interpolation}} f \xrightarrow{\text{resampling}} \{v_{\mathbf{k}}\} \in \mathbb{R}^{N_1 \times \dots \times N_d}$$

Suppose the interpolating filter φ is $(M_1, \dots, M_d)$ -periodic, i.e. for any $1 \leq i \leq d$, $\varphi(\dots, x_i, \dots) = \varphi(\dots, x_i + M_i, \dots)$. $\mathbf{u}$ is interpolated to function $f: \mathbb{R}^d \rightarrow \mathbb{R}^d$ by

$$f(x_1, \dots, x_d) = \sum_{k_1=0}^{M_1-1} \dots \sum_{k_d=0}^{M_d-1} u_{k_1, \dots, k_d} \varphi(x_1 - k_1, \dots, x_d - k_d) \quad (\text{S7})$$

f is discretized evenly on $[0, M_1] \times \dots \times [0, M_d]$ to $\mathbf{v} = \{v_{k_1, \dots, k_d}\}$ by taking $v_{k_1, \dots, k_d} = f(k_1 \frac{M_1}{N_1}, \dots, k_d \frac{M_d}{N_d})$.

Theorem. Suppose the interpolating filter φ is $(M_1, \dots, M_d)$ -periodic, i.e. for any $1 \leq i \leq d$, $\varphi(\dots, x_i, \dots) = \varphi(\dots, x_i + M_i, \dots)$. Denote the DFTs of $\mathbf{u}$ and $\mathbf{v}$ by $\{\tilde{u}_{\mathbf{m}}\} \in \mathbb{C}^{M_1 \times \dots \times M_d}$ with $\mathbf{m} = (m_1, \dots, m_d)$ and $\{\tilde{v}_{\mathbf{n}}\} \in \mathbb{C}^{N_1 \times \dots \times N_d}$ with $\mathbf{n} = (n_1, \dots, n_d)$, respectively. Denote the Fourier series of φ and f by $\{c_{\mathbf{p}}(\varphi)\}_{\mathbf{p} \in \mathbb{Z}^d}$ and $\{c_{\mathbf{p}}(f)\}_{\mathbf{p} \in \mathbb{Z}^d}$. Then we have

$$\tilde{v}_{\mathbf{n}} = \left(\prod_{i=1}^d M_i \right) \cdot \sum_{\mathbf{z} \in \mathbb{Z}^d} c_{\mathbf{z} \odot \mathbf{N} + \mathbf{n}}(\varphi) \cdot \tilde{u}_{[(\mathbf{z} \odot \mathbf{N} + \mathbf{n}) \bmod \mathbf{M}]} \quad (\text{S8})$$

where $\mathbf{M} = (M_1, \dots, M_d)$ and $\mathbf{N} = (N_1, \dots, N_d)$. $\bmod$ and $\odot$ is element-wise modulo and multiplication, respectively.

Proof. The proof is similar to case of 1-dimension. By the definitions of the multidimensional DFT and Fourier series, we have

$$u_{k_1, \dots, k_d} = \sum_{m_1=0}^{M_1-1} \left(\omega_{M_1}^{k_1 m_1} \sum_{m_2=0}^{M_2-1} \left(\omega_{M_2}^{k_2 m_2} \dots \right. \right. \quad (\text{S9})$$

$$\left. \sum_{m_d=0}^{M_d-1} \omega_{M_d}^{k_d m_d} \cdot \tilde{u}_{m_1, \dots, m_d} \right) \quad (\text{S10})$$

$$v_{k_1, \dots, k_d} = \sum_{n_1=0}^{N_1-1} \left(\omega_{N_1}^{k_1 n_1} \sum_{n_2=0}^{N_2-1} \left(\omega_{N_2}^{k_2 n_2} \dots \right. \right. \quad (\text{S11})$$

$$\left. \sum_{n_d=0}^{N_d-1} \omega_{N_d}^{k_d n_d} \cdot \tilde{v}_{n_1, \dots, n_d} \right) \quad (\text{S12})$$

$$\varphi(x_1, \dots, x_d) = \sum_{p_1 \in \mathbb{Z}} \left(\omega_{M_1}^{x_1 p_1} \sum_{p_2 \in \mathbb{Z}} \left(\omega_{M_2}^{x_2 p_2} \dots \right. \right. \quad (\text{S13})$$

$$\left. \sum_{p_d \in \mathbb{Z}} \omega_{M_d}^{x_d p_d} \cdot c_{p_1, \dots, p_d}(\varphi) \right) \quad (\text{S14})$$

$$f(x_1, \dots, x_d) = \sum_{p_1 \in \mathbb{Z}} \left(\omega_{M_1}^{x_1 p_1} \sum_{p_2 \in \mathbb{Z}} \left(\omega_{M_2}^{x_2 p_2} \dots \right. \right. \quad (\text{S15})$$

$$\left. \sum_{p_d \in \mathbb{Z}} \omega_{M_d}^{x_d p_d} \cdot c_{p_1, \dots, p_d}(f) \right) \quad (\text{S16})$$

Or by setting $\omega_{\mathbf{M}}^{\mathbf{x}} := e^{2\pi i(\mathbf{x}^T \cdot \frac{1}{\mathbf{M}})}$, $\mathcal{M} := \llbracket 0, M_1 - 1 \rrbracket \times \dots \times \llbracket 0, M_d - 1 \rrbracket$ and $\mathcal{N} := \llbracket 0, N_1 - 1 \rrbracket \times \dots \times \llbracket 0, N_d - 1 \rrbracket$, we have

$$u_{\mathbf{k}} = \sum_{\mathbf{m} \in \mathcal{M}} \tilde{u}_{\mathbf{m}} \cdot \omega_{\mathbf{M}}^{\mathbf{k} \odot \mathbf{m}} \quad (\text{S17})$$

$$v_{\mathbf{k}} = \sum_{\mathbf{n} \in \mathcal{N}} \tilde{v}_{\mathbf{n}} \cdot \omega_{\mathbf{N}}^{\mathbf{k} \odot \mathbf{n}} \quad (\text{S18})$$

$$\varphi(\mathbf{x}) = \sum_{\mathbf{p} \in \mathbb{Z}^d} c_{\mathbf{p}}(\varphi) \cdot \omega_{\mathbf{M}}^{\mathbf{x} \odot \mathbf{p}} \quad (\text{S19})$$

$$f(\mathbf{x}) = \sum_{\mathbf{p} \in \mathbb{Z}^d} c_{\mathbf{p}}(f) \cdot \omega_{\mathbf{M}}^{\mathbf{x} \odot \mathbf{p}} \quad (\text{S20})$$

The interpolation process can be rewritten by the coefficients of DFT and Fourier series:

$$\begin{aligned} f(\mathbf{x}) &= \sum_{\mathbf{k} \in \mathcal{M}} u_{\mathbf{k}} \varphi(\mathbf{x} - \mathbf{k}) \\ &= \sum_{\mathbf{k} \in \mathcal{M}} \left(\sum_{\mathbf{m} \in \mathcal{M}} \tilde{u}_{\mathbf{m}} \cdot \omega_{\mathbf{M}}^{\mathbf{k} \odot \mathbf{m}} \right) \cdot \left(\sum_{\mathbf{p} \in \mathbb{Z}^d} c_{\mathbf{p}}(\varphi) \cdot \omega_{\mathbf{M}}^{(\mathbf{x} - \mathbf{k}) \odot \mathbf{p}} \right) \\ &= \sum_{\mathbf{k} \in \mathcal{M}} \sum_{\mathbf{m} \in \mathcal{M}} \sum_{\mathbf{p} \in \mathbb{Z}^d} \tilde{u}_{\mathbf{m}} \cdot \omega_{\mathbf{M}}^{\mathbf{k} \odot \mathbf{m}} \cdot c_{\mathbf{p}}(\varphi) \cdot \omega_{\mathbf{M}}^{(\mathbf{x} - \mathbf{k}) \odot \mathbf{p}} \\ &= \sum_{\mathbf{p} \in \mathbb{Z}^d} \omega_{\mathbf{M}}^{\mathbf{x} \odot \mathbf{p}} \underbrace{\sum_{\mathbf{k} \in \mathcal{M}} \sum_{\mathbf{m} \in \mathcal{M}} \tilde{u}_{\mathbf{m}} \cdot \omega_{\mathbf{M}}^{\mathbf{k} \odot \mathbf{m}} \cdot c_{\mathbf{p}}(\varphi) \cdot \omega_{\mathbf{M}}^{-\mathbf{k} \odot \mathbf{p}}}_{c_{\mathbf{p}}(f)} \end{aligned} \quad (\text{S21})$$

By identification from Eq. S20 we have

$$\begin{aligned} c_{\mathbf{p}}(f) &= \sum_{\mathbf{k} \in \mathcal{M}} \sum_{\mathbf{m} \in \mathcal{M}} \tilde{u}_{\mathbf{m}} \cdot \omega_{\mathbf{M}}^{\mathbf{k} \odot \mathbf{m}} \cdot c_{\mathbf{p}}(\varphi) \cdot \omega_{\mathbf{M}}^{-\mathbf{k} \odot \mathbf{p}} \\ &= c_{\mathbf{p}}(\varphi) \sum_{\mathbf{m} \in \mathcal{M}} \tilde{u}_{\mathbf{m}} \sum_{\mathbf{k} \in \mathcal{M}} \omega_{\mathbf{M}}^{\mathbf{k} \odot (\mathbf{m} - \mathbf{p})} \end{aligned} \quad (\text{S22})$$

As $\mathbf{p} \in \mathbb{Z}^d$ and $\mathbf{m} \in \llbracket 0, M_1 - 1 \rrbracket \times \dots \times \llbracket 0, M_d - 1 \rrbracket$, $\sum_{\mathbf{k} \in \mathcal{M}} \omega_{\mathbf{M}}^{\mathbf{k} \odot (\mathbf{m} - \mathbf{p})} = \left(\prod_{i=1}^d M_i \right) \mathbb{1}_{\{\mathbf{p} \bmod \mathbf{M} = \mathbf{m}\}}$. Indeed,

$$\begin{aligned} \sum_{\mathbf{k} \in \mathcal{M}} \omega_{\mathbf{M}}^{\mathbf{k} \odot (\mathbf{m} - \mathbf{p})} &= \sum_{k_1=0}^{M_1-1} \dots \sum_{k_d=0}^{M_d-1} \omega_{M_1}^{k_1(m_1 - p_1)} \dots \omega_{M_d}^{k_d(m_d - p_d)} \\ &= \left(\sum_{k_1=0}^{M_1-1} \omega_{M_1}^{k_1(m_1 - p_1)} \right) \dots \left(\sum_{k_d=0}^{M_d-1} \omega_{M_d}^{k_d(m_d - p_d)} \right) \\ &= M_1 \mathbb{1}_{\{p_1 \bmod M_1 = m_1\}} \dots M_d \mathbb{1}_{\{p_d \bmod M_d = m_d\}} \\ &= \left(\prod_{i=1}^d M_i \right) \mathbb{1}_{\{\mathbf{p} \bmod \mathbf{M} = \mathbf{m}\}} \end{aligned} \quad (\text{S23})$$

Eq. S22 becomes

$$\begin{aligned} c_{\mathbf{p}}(f) &= c_{\mathbf{p}}(\varphi) \sum_{\mathbf{m} \in \mathcal{M}} \tilde{u}_{\mathbf{m}} \left(\prod_{i=1}^d M_i \right) \mathbb{1}_{\{\mathbf{p} \bmod \mathbf{M} = \mathbf{m}\}} \\ &= \left(\prod_{i=1}^d M_i \right) c_{\mathbf{p}}(\varphi) \cdot \tilde{u}_{[\mathbf{p} \bmod \mathbf{M}]} \end{aligned} \quad (\text{S24})$$

Similarly to the proof for 1-dimension, we now prove the relation between $\{\tilde{v}_n\}$ and $\{c_p(f)\}$. Given the resampling by $v_k = f(\mathbf{k} \odot \mathbf{M} \odot \frac{1}{N})$, we have

$$\begin{aligned}
v_k &= f(\mathbf{k} \odot \mathbf{M} \odot \frac{1}{N}) \\
&= \sum_{\mathbf{p} \in \mathbb{Z}^d} c_p(f) \cdot \omega_{\mathbf{M}}^{\mathbf{k} \odot \mathbf{M} \odot \frac{1}{N} \odot \mathbf{p}} \\
&= \sum_{\mathbf{p} \in \mathbb{Z}^d} c_p(f) \cdot e^{2\pi i \left((\mathbf{k} \odot \mathbf{M} \odot \frac{1}{N} \odot \mathbf{p})^\top \cdot \frac{1}{\mathbf{M}} \right)} \\
&= \sum_{\mathbf{p} \in \mathbb{Z}^d} c_p(f) \cdot e^{2\pi i \left((\mathbf{k} \odot \mathbf{p})^\top \cdot \frac{1}{N} \right)} \\
&= \sum_{\mathbf{p} \in \mathbb{Z}^d} c_p(f) \cdot \omega_N^{\mathbf{k} \odot \mathbf{p}} \\
&\quad (\text{split } \mathbb{Z}^d \text{ into disjoint hyper-cubes of size } N_1 \times N_2 \times \dots \\
&\quad \quad \quad \times N_d) \\
&= \sum_{\mathbf{z} \in \mathbb{Z}^d} \sum_{\mathbf{n} \in \mathcal{N}} c_{\mathbf{z} \odot \mathbf{N} + \mathbf{n}}(f) \cdot \omega_N^{\mathbf{k} \odot (\mathbf{z} \odot \mathbf{N} + \mathbf{n})} \\
&= \sum_{\mathbf{n} \in \mathcal{N}} \underbrace{\sum_{\mathbf{z} \in \mathbb{Z}^d} c_{\mathbf{z} \odot \mathbf{N} + \mathbf{n}}(f) \cdot \omega_N^{\mathbf{k} \odot \mathbf{n}}}_{\tilde{v}_n}
\end{aligned} \tag{S25}$$

By identification from Eq. S18 we have

$$\begin{aligned}
\tilde{v}_n &= \sum_{\mathbf{z} \in \mathbb{Z}^d} c_{\mathbf{z} \odot \mathbf{N} + \mathbf{n}}(f) \\
&= \left(\prod_{i=1}^d M_i \right) \sum_{\mathbf{z} \in \mathbb{Z}^d} c_{\mathbf{z} \odot \mathbf{N} + \mathbf{n}}(\varphi) \cdot \tilde{u}_{[(\mathbf{z} \odot \mathbf{N} + \mathbf{n}) \bmod \mathbf{M}]} \tag{S26} \\
&\triangleq cst \cdot \sum_{\mathbf{z} \in \mathbb{Z}^d} c_{\mathbf{z} \odot \mathbf{N} + \mathbf{n}}(\varphi) \cdot \tilde{u}_{[(\mathbf{z} \odot \mathbf{N} + \mathbf{n}) \bmod \mathbf{M}]}
\end{aligned}$$

with $cst \triangleq \prod_{i=1}^d M_i$ a constant independent of the coefficient entry $\mathbf{n}$.

□

APPENDIX C

COVARIANCE BETWEEN TWO ARBITRARY COEFFICIENTS IN A SPECTRUM OF RESAMPLED WHITE NOISE

Theorem. Let the original signal $\{u_k\} \in \mathbb{R}^M$ be a sequence of white noise with $\text{Var}[u_k] = \sigma^2$ and $\text{E}[u_k] = 0$, which is resampled to $\{v_k\} \in \mathbb{R}^N$ using an M -periodic interpolating filter $\varphi: \mathbb{R} \mapsto \mathbb{R}$. Let $\tilde{v}_m$ and $\tilde{v}_n$ be two arbitrary frequency components in the DFT of the resampled signal, $m, n \in \mathbb{Z}$. Denote by $L = \text{lcm}(M, N)$ and $\text{gcd}(M, N)$ the least common multiple and the greatest common divisor of M and N , respectively. We have $\text{Cov}[\tilde{v}_n, \tilde{v}_m]$ given by:

- if $m - n$ is not a multiple of $\text{gcd}(M, N)$, $\text{Cov}[\tilde{v}_n, \tilde{v}_m] = 0$;
- if $m - n$ is a multiple of $\text{gcd}(M, N)$, define $\mathcal{K}(x)$ as the solution of

$$\begin{aligned}
&\min_{k'} |k'| \\
&\text{s.t. } k'M \pmod{N} = x \pmod{N} \\
&k' \in \mathbb{Z},
\end{aligned} \tag{S27}$$

then the covariance is given by

$$\begin{aligned}
&\text{Cov}[\tilde{v}_n, \tilde{v}_m] \\
&= M\sigma^2 \sum_{q \in \mathbb{Z}} \sum_{z \in \mathbb{Z}} c_{zN+n}(\varphi) \cdot \overline{c_{zN+n+kM+qL}(\varphi)}
\end{aligned} \tag{S28}$$

where $k = \mathcal{K}(m - n)$.

Proof. It is easy to show $\text{E}[\tilde{u}_n] = 0$ and $\text{Var}[\tilde{u}_n] = \text{E}[\tilde{u}_n \tilde{u}_n] = \frac{\sigma^2}{M}$ for any $n \in \mathbb{Z}$. According to Theorem 1 we have

$$\begin{aligned}
\tilde{v}_n &= M \sum_{z \in \mathbb{Z}} c_{zN+n}(\varphi) \cdot \tilde{u}_{zN+n} \\
\tilde{v}_m &= M \sum_{z \in \mathbb{Z}} c_{zN+m}(\varphi) \cdot \tilde{u}_{zN+m}
\end{aligned} \tag{S29}$$

Naturally, $\text{E}[\tilde{v}_m] = \text{E}[\tilde{v}_n] = 0$. Their covariance is

$$\begin{aligned}
&\text{Cov}[\tilde{v}_n, \tilde{v}_m] \\
&= M^2 \text{E} \left[\left(\sum_{z \in \mathbb{Z}} c_{zN+n}(\varphi) \cdot \tilde{u}_{zN+n} \right) \overline{\left(\sum_{z' \in \mathbb{Z}} c_{z'N+m}(\varphi) \cdot \tilde{u}_{z'N+m} \right)} \right] \\
&= M^2 \sum_{z \in \mathbb{Z}} \sum_{z' \in \mathbb{Z}} c_{zN+n}(\varphi) \overline{c_{z'N+m}(\varphi)} \cdot \text{E}[\tilde{u}_{zN+n} \tilde{u}_{z'N+m}].
\end{aligned} \tag{S30}$$

As the DFT $\{\tilde{u}_n\}$ is M -periodic, $\text{E}[\tilde{u}_{zN+n} \tilde{u}_{z'N+m}] = \frac{\sigma^2}{M}$ if and only if $\exists k' \in \mathbb{Z}$ such that

$$\begin{aligned}
&zN + n + k'M = z'N + m \\
&\Leftrightarrow (z - z')N + k'M = m - n,
\end{aligned} \tag{S31}$$

Eq. S31 is actually a linear Diophantine equation $Nx + My = m - n$ with x and y the integer unknowns. Therefore, if $m - n$ is not a multiple of $\text{gcd}(M, N)$, Eq. S31 has no solution and $\text{E}[\tilde{u}_{zN+n} \tilde{u}_{z'N+m}] = 0$ for all the $z, z' \in \mathbb{Z}$, which yields $\text{Cov}[\tilde{v}_n, \tilde{v}_m] = 0$.

If $m - n$ is a multiple of $\text{gcd}(M, N)$, then the possible k' that make Eq. S31 hold satisfy $m - n \pmod{N} = k'M \pmod{N}$. Let $\mathbf{K}$ be the set of the possible values of k' and let $k = \arg \min_{k' \in \mathbf{K}} |k'|$, then the possible $z - z'$ that make Eq. S31

hold are $z - z' \in \{\frac{m-n-kM}{N} + q\frac{L}{N}, q \in \mathbb{Z}\}$. Finally, Eq. S30 is rearranged to be

$$\begin{aligned}
& \text{Cov}[\tilde{v}_n, \tilde{v}_m] \\
&= M^2 \sum_{z \in \mathbb{Z}} \sum_{q \in \mathbb{Z}} c_{zN+n}(\varphi) \overline{c_{z - (\frac{m-n-kM}{N} + q\frac{L}{N})N+m}(\varphi)} \cdot \frac{\sigma^2}{M} \\
&= M\sigma^2 \sum_{q \in \mathbb{Z}} \sum_{z \in \mathbb{Z}} c_{zN+n}(\varphi) \overline{c_{zN+n+kM+qL}(\varphi)}
\end{aligned} \tag{S32}$$

□

APPENDIX D

APPENDIX: MAXIMUM OF CORRELATION

Theorem 4. Let $M, N \in \mathbb{N}_+^*$ and $n \in \mathbb{Z}$. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be two M -periodic functions. Denote the Fourier series of f and g by $\{c_z(f)\}_{z \in \mathbb{Z}}$ and $\{c_z(g)\}_{z \in \mathbb{Z}}$, respectively. Denote $\omega_M = e^{\frac{2\pi i}{M}}$. If $c_z(g) = c_{zN+n}(f)$ for any $z \in \mathbb{Z}$, then for any $x \in \mathbb{R}$, the relationship between f and g is given by

$$g(x) = \frac{1}{N} \omega_M^{-n\frac{x}{N}} \sum_{p=0}^{N-1} f\left(\frac{x}{N} + p\frac{M}{N}\right) \omega_N^{-np} \tag{S33}$$

Proof. By definition, the Fourier series of f is

$$c_z(g) = \frac{1}{M} \int_0^M g(x) \omega_M^{-zx} dx \tag{S34}$$

And the expansion of $c_{zN+n}(f)$ is

$$\begin{aligned}
& c_{zN+n}(f) \\
&= \frac{1}{M} \int_0^M f(x) \omega_M^{-(zN+n)x} dx \\
& \text{(set } t = Nx) \\
&= \frac{1}{MN} \int_0^{MN} f\left(\frac{t}{N}\right) \omega_M^{-(zN+n)\frac{t}{N}} dt \\
&= \frac{1}{MN} \sum_{p=0}^{N-1} \int_{pM}^{(p+1)M} f\left(\frac{t}{N}\right) \omega_M^{-(zN+n)\frac{t}{N}} dt \\
& \text{(set } t' = t - pM \text{ for each term of } p) \\
&= \frac{1}{MN} \sum_{p=0}^{N-1} \int_0^M f\left(\frac{t'}{N} + p\frac{M}{N}\right) \omega_M^{-n\frac{t'}{N}} \omega_N^{-np} \omega_M^{-zt'} dt' \\
&= \frac{1}{M} \int_0^M \underbrace{\frac{1}{N} \sum_{p=0}^{N-1} f\left(\frac{t'}{N} + p\frac{M}{N}\right) \omega_M^{-n\frac{t'}{N}} \omega_N^{-np}}_{g(t')} \omega_M^{-zt'} dt'
\end{aligned} \tag{S35}$$

By identification from Eq. S35 and Eq. S34 we obtain

$$g(x) = \frac{1}{N} \omega_M^{-n\frac{x}{N}} \sum_{p=0}^{N-1} f\left(\frac{x}{N} + p\frac{M}{N}\right) \omega_N^{-np} \tag{S36}$$

□

One can show that the constructed g in Eq. S33 is M -periodic.

Theorem 5. Let $M, N \in \mathbb{N}_+^*$ and $k \in \mathbb{Z}$. Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be an M -periodic real-valued function whose Fourier series is $\{c_z(\varphi)\}_{z \in \mathbb{Z}}$. Then the function $r(n) = \sum_{z \in \mathbb{Z}} c_{zN+n}(\varphi) \cdot \overline{c_{zN+n+kM}(\varphi)}$ for $n \in \mathbb{Z}$ can be expressed in the time domain by:

$$r(n) = \frac{1}{MN} \sum_{\delta=0}^{N-1} r_\delta(n) \tag{S37}$$

where $r_\delta(n) = \cos\left(\frac{2\pi\delta}{N}\left(\frac{kM}{2} + n\right)\right) \int_0^M \varphi\left(t + \frac{\delta}{2}\frac{M}{N}\right) \varphi\left(t - \frac{\delta}{2}\frac{M}{N}\right) e^{i2\pi kt} dt$. Moreover, $n^* = -\frac{kM}{2}$ is a critical point of $r(n)$.

Proof. We start with calculating $r(n) = \sum_{z \in \mathbb{Z}} c_{zN+n}(\varphi) \cdot \overline{c_{zN+n+kM}(\varphi)}$. Let $A : \mathbb{R} \mapsto \mathbb{R}$ be an M -periodic function whose Fourier series $\{c_z(A)\}_{z \in \mathbb{Z}}$ satisfies $c_z(A) = c_{zN+n}(\varphi)$. According to Theorem 4,

$$A(x) = \frac{1}{N} \sum_{p=0}^{N-1} \varphi\left(\frac{x}{N} + p\frac{M}{N}\right) \omega_M^{-n\frac{x}{N}} \omega_N^{-np} \quad (\text{S38})$$

Likewise, let $B : \mathbb{R} \mapsto \mathbb{R}$ be an M -periodic function whose Fourier series $\{c_z(B)\}_{z \in \mathbb{Z}}$ satisfies $c_z(B) = c_{zN+n+kM}(\varphi)$, then

$$B(x) = \frac{1}{N} \sum_{p=0}^{N-1} \varphi\left(\frac{x}{N} + p\frac{M}{N}\right) \omega_M^{-(n+kM)\frac{x}{N}} \omega_N^{-(n+kM)p} \quad (\text{S39})$$

According to the Parseval's theorem, $\sum_{z \in \mathbb{Z}} c_z(A) \overline{c_z(B)} = \frac{1}{M} \int_0^M A(x) \overline{B(x)} dx$, it suffices to compute the integral on the right-hand side to compute $r(n)$.

$$\begin{aligned} r(n) &= \frac{1}{M} \int_0^M A(x) \overline{B(x)} dx \\ &= \frac{1}{MN^2} \int_0^M \sum_{p=0}^{N-1} \sum_{q=0}^{N-1} \varphi\left(\frac{x}{N} + p\frac{M}{N}\right) \overline{\varphi\left(\frac{x}{N} + q\frac{M}{N}\right)} \dots \\ &\quad \cdot \omega_N^{kx} \omega_N^{n(q-p)+kMq} dx \\ &= \frac{1}{MN^2} \sum_{p=0}^{N-1} \sum_{q=0}^{N-1} \int_0^M \varphi\left(\frac{x}{N} + p\frac{M}{N}\right) \overline{\varphi\left(\frac{x}{N} + q\frac{M}{N}\right)} \dots \\ &\quad \cdot \omega_N^{k(x+qM)+n(q-p)} dx \\ &= \frac{1}{MN^2} \sum_{p=0}^{N-1} \sum_{q=0}^{N-1} \beta_{p,q}(n) \end{aligned} \quad (\text{S40})$$

where

$$\beta_{p,q}(n) = \int_0^M \varphi\left(\frac{x}{N} + p\frac{M}{N}\right) \overline{\varphi\left(\frac{x}{N} + q\frac{M}{N}\right)} \omega_N^{k(x+qM)+n(q-p)} dx \quad (\text{S41})$$

Note that $\beta_{p,q}(n) = \beta_{p,q+N}(n)$, Eq. S40 can be rewritten as

$$\begin{aligned} r(n) &= \frac{1}{MN^2} \sum_{p=0}^{N-1} \sum_{q=p}^{p+N-1} \beta_{p,q}(n) \\ &= \frac{1}{MN^2} \sum_{p=0}^{N-1} \sum_{\delta=0}^{N-1} \beta_{p,p+\delta}(n) \\ &= \frac{1}{MN^2} \sum_{\delta=0}^{N-1} \sum_{p=0}^{N-1} \beta_{p,p+\delta}(n) \end{aligned} \quad (\text{S42})$$

with $\delta = q - p$. Similarly, Eq. S40 can be rewritten as

$$r(n) = \frac{1}{MN^2} \sum_{\delta=0}^{N-1} \sum_{p=0}^{N-1} \beta_{p,p-\delta}(n) \quad (\text{S43})$$

Averaging Eq. S42 and Eq. S43 amounts to

$$r(n) = \frac{1}{2MN^2} \sum_{\delta=0}^{N-1} \sum_{p=0}^{N-1} (\beta_{p,p-\delta}(n) + \beta_{p,p+\delta}(n)) \quad (\text{S44})$$

We now focus on calculating the sum for each fixed δ . From Eq. S41 we have

$$\begin{aligned} &\sum_{p=0}^{N-1} \beta_{p,p+\delta}(n) \\ &= \sum_{p=0}^{N-1} \int_0^M \varphi\left(\frac{x}{N} + p\frac{M}{N}\right) \overline{\varphi\left(\frac{x}{N} + (p+\delta)\frac{M}{N}\right)} \dots \\ &\quad \cdot \omega_N^{k(x+(p+\delta)M)+n\delta} dx \\ &\quad (\text{set } t = \frac{x}{N} + (p + \frac{\delta}{2})\frac{M}{N}) \\ &= N \sum_{p=0}^{N-1} \int_{(p+\frac{\delta}{2})\frac{M}{N}}^{(p+\frac{\delta}{2}+1)\frac{M}{N}} \varphi\left(t - \frac{\delta}{2}\frac{M}{N}\right) \overline{\varphi\left(t + \frac{\delta}{2}\frac{M}{N}\right)} \dots \\ &\quad \cdot \omega_N^{k(Nt+\frac{\delta}{2}M)+n\delta} dt \\ &= N \int_{\frac{\delta}{2}\frac{M}{N}}^{(N+\frac{\delta}{2})\frac{M}{N}} \varphi\left(t - \frac{\delta}{2}\frac{M}{N}\right) \overline{\varphi\left(t + \frac{\delta}{2}\frac{M}{N}\right)} \omega_N^{k(Nt+\frac{\delta}{2}M)+n\delta} dt \\ &= N \int_{\frac{\delta}{2}\frac{M}{N}}^M \varphi\left(t - \frac{\delta}{2}\frac{M}{N}\right) \overline{\varphi\left(t + \frac{\delta}{2}\frac{M}{N}\right)} \omega_N^{k(Nt+\frac{\delta}{2}M)+n\delta} dt \\ &\quad + N \int_M^{M+\frac{\delta}{2}\frac{M}{N}} \varphi\left(t - \frac{\delta}{2}\frac{M}{N}\right) \overline{\varphi\left(t + \frac{\delta}{2}\frac{M}{N}\right)} \omega_N^{k(Nt+\frac{\delta}{2}M)+n\delta} dt \end{aligned} \quad (\text{S45})$$

As $t \mapsto \varphi\left(t - \frac{\delta}{2}\frac{M}{N}\right) \overline{\varphi\left(t + \frac{\delta}{2}\frac{M}{N}\right)} \omega_N^{k(Nt+\frac{\delta}{2}M)+n\delta}$ is M -periodic,

$$\begin{aligned} &\sum_{p=0}^{N-1} \beta_{p,p+\delta}(n) \\ &= N \int_{\frac{\delta}{2}\frac{M}{N}}^M \varphi\left(t - \frac{\delta}{2}\frac{M}{N}\right) \overline{\varphi\left(t + \frac{\delta}{2}\frac{M}{N}\right)} \omega_N^{k(Nt+\frac{\delta}{2}M)+n\delta} dt \\ &\quad + N \int_0^{\frac{\delta}{2}\frac{M}{N}} \varphi\left(t - \frac{\delta}{2}\frac{M}{N}\right) \overline{\varphi\left(t + \frac{\delta}{2}\frac{M}{N}\right)} \omega_N^{k(Nt+\frac{\delta}{2}M)+n\delta} dt \\ &= N \int_0^M \varphi\left(t - \frac{\delta}{2}\frac{M}{N}\right) \overline{\varphi\left(t + \frac{\delta}{2}\frac{M}{N}\right)} \omega_N^{k(Nt+\frac{\delta}{2}M)+n\delta} dt \end{aligned} \quad (\text{S46})$$

Likewise,

$$\begin{aligned} &\sum_{p=0}^{N-1} \beta_{p,p-\delta}(n) \\ &= N \int_0^M \varphi\left(t + \frac{\delta}{2}\frac{M}{N}\right) \overline{\varphi\left(t - \frac{\delta}{2}\frac{M}{N}\right)} \omega_N^{k(Nt-\frac{\delta}{2}M)-n\delta} dt \end{aligned} \quad (\text{S47})$$

Combining Eq. S46 with Eq. S47 amounts to

$$\begin{aligned} &\sum_{p=0}^{N-1} \beta_{p,p+\delta}(n) + \beta_{p,p-\delta}(n) \\ &= N \int_0^M \varphi\left(t + \frac{\delta}{2}\frac{M}{N}\right) \overline{\varphi\left(t - \frac{\delta}{2}\frac{M}{N}\right)} \omega_N^{kNt} \dots \\ &\quad \cdot (\omega_N^{\frac{\delta}{2}kM+n\delta} + \omega_N^{-\frac{\delta}{2}kM-n\delta}) dt \\ &= 2N \cdot \cos\left(\frac{2\pi\delta}{N}\left(\frac{kM}{2} + n\right)\right) \dots \\ &\quad \cdot \int_0^M \varphi\left(t + \frac{\delta}{2}\frac{M}{N}\right) \overline{\varphi\left(t - \frac{\delta}{2}\frac{M}{N}\right)} e^{i2\pi kt} dt \end{aligned} \quad (\text{S48})$$

Combining Eq. S44 with Eq. S48, we finally obtain

$$r(n) = \frac{1}{MN} \sum_{\delta=0}^{N-1} r_{\delta}(n) \quad (\text{S49})$$

where $r_{\delta}(n) := \cos(\frac{2\pi\delta}{N}(\frac{kM}{2} + n)) \int_0^M \varphi(t + \frac{\delta}{2}\frac{M}{N})\varphi(t - \frac{\delta}{2}\frac{M}{N})e^{i2\pi kt}dt$.

We remark that $r_{\delta=0}(n) = \int_0^M e^{i2\pi kt}\varphi(t)^2dt$ is a constant independent of n , and $n^* = -\frac{kM}{2}$ is a critical point of $r_{\delta}(n)$ for $\delta \neq 0$. Thus $n^* = -\frac{kM}{2}$ is a critical point of $r(n)$. $\square$

Corollary 1. *For a common interpolating filter φ which is even, $\int_0^M \varphi(t + \frac{\delta}{2}\frac{M}{N})\varphi(t - \frac{\delta}{2}\frac{M}{N})e^{i2\pi kt}dt \in \mathbb{R}$. If we admit $\int_0^M \varphi(t + \frac{\delta}{2}\frac{M}{N})\varphi(t - \frac{\delta}{2}\frac{M}{N})e^{i2\pi kt}dt \geq 0$ for any $\delta \in \llbracket 0, N-1 \rrbracket$, then $r(n)$ is maximized at $n^* = -\frac{kM}{2}$. The maximum of $r(n)$ is*

$$\max_n r(n) = \frac{1}{MN} \sum_{\delta=0}^{N-1} \int_0^M \varphi(t + \frac{\delta}{2}\frac{M}{N})\varphi(t - \frac{\delta}{2}\frac{M}{N})e^{i2\pi kt}dt \quad (\text{S50})$$

Corollary 2. *If φ is the nearest-neighbor filter and if $M \geq N$, then $r(n) = 0$.*

Proof. Indeed, the nearest-neighbor filter is expressed by $\varphi(t) = \mathbb{1}_{[-1/2, 1/2]}(t)$. When $\delta = 0$,

$$\int_0^M \varphi(t + \frac{\delta}{2}\frac{M}{N})\varphi(t - \frac{\delta}{2}\frac{M}{N})e^{i2\pi kt}dt = \int_{-1/2}^{1/2} 1 \cdot e^{i2\pi kt}dt = 0. \quad (\text{S51})$$

When $\delta \in \llbracket 1, N-1 \rrbracket$, $\int_0^M \varphi(t + \frac{\delta}{2}\frac{M}{N})\varphi(t - \frac{\delta}{2}\frac{M}{N})e^{i2\pi kt}dt = 0$ because $\delta \frac{M}{N} \geq 1$ and $\varphi(t + \frac{\delta}{2}\frac{M}{N})\varphi(t - \frac{\delta}{2}\frac{M}{N}) = 0$ for any t .

Therefore, for any $\delta \in \llbracket 0, N-1 \rrbracket$ we have $r_{\delta}(n) = 0$, resulting in $r(n) = 0$. $\square$

Corollary 2 describes the fact that any two frequency components in the spectrum of downsampled ($M \geq N$) white noise are uncorrelated when the nearest-neighbor filter is used for downsampling. Indeed, a white noise after downsampling using the nearest-neighbor filter is simply a subsequence of the original white noise, which is still a white noise. Hence, the frequency components in the output spectrum are uncorrelated.

However, in the case of downsampling a signal whose power spectrum density is not homogeneous using the nearest-neighbor filter, there exist pairs of frequency components again in the spectrum of the downsampled signal that are correlated. Fig. 12 showing very detecting accuracies on images downsampled with the nearest-neighbor filter also confirms this fact.

APPENDIX E

THEOREM: SPECTRUM FOLDING OF NON-ALIGNED RESAMPLING

The previous theorems describe the spectrum folding of resampling a discrete signal from the entire interpolated function. In practice, however, it is common to process the resampling on an interval in a length that is different from the period of the interpolated function. For instance, sometimes the corner points are required to be aligned³ when resampling a signal. In such a case, the interval of discretizing the interpolated function is not exactly in alignment with the support of the interpolated function. This would slightly alter the phenomenon of spectrum folding and change the distance between correlated frequency components in the spectrum.

Consider again a 1D signal $\{u_k\} \in \mathbb{R}^M$ of M samples and its resampled version $\{v_k\} \in \mathbb{R}^N$, with $M, N \in \mathbb{N}^*$ being any two positive integers. The non-aligned resampling is modeled in two steps: first the samples $\{u_k\}$ are interpolated to a continuous function f using an M -periodic interpolating filter φ in the same way as the aligned-resampling, then f is resampled to the discrete signal $\{v_k\}$ of N samples on a support in length M' that is slightly different from M .

Again, the continuous function $f: \mathbb{R} \rightarrow \mathbb{R}$ as the output of the interpolation is $f(x) = \sum_{k=0}^{M-1} u_k \varphi(x - k)$.

For the non-aligned resampling, let the size of the resampling interval be $M' = M + \Delta$ with $\Delta \in [-1, 1]$, and f is discretized evenly on $[0, M']$ to sequence $\mathbf{v} = \{v_k\} \in \mathbb{R}^N$ by taking $v_k = f(k \frac{M'}{N})$. Then in the spectrum of the resampled signal, two frequency components separated by $M-1$, M or $M+1$ components are correlated.

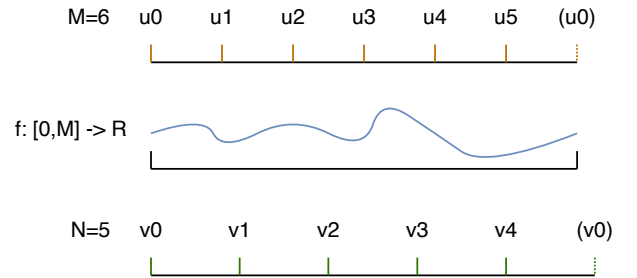


Fig. S2: Non-aligned resampling: the intervals for the original sequence and the resampled sequence are slightly different.

Theorem 6. *Suppose the interpolating filter φ is M -periodic. Denote the DFTs of $\{u_k\}$ and $\{v_k\}$ by $\{\tilde{u}_m\} \in \mathbb{C}^M$ and $\{\tilde{v}_n\} \in \mathbb{C}^N$, respectively. Denote the Fourier coefficients of φ and f by $\{c_p(\varphi)\}_p$ and $\{c_p(f)\}_p$, respectively.*

Define a 2D series $T_{p,q}$ by

$$T_{p,q} = -ie^{\pi i M'(\frac{q}{M} - \frac{p}{M'})} \text{sinc}\left(\pi M' \left(\frac{q}{M} - \frac{p}{M'}\right)\right).$$

Then:

- if $\Delta = 0$, we have

$$\tilde{v}_n = M \sum_{z \in \mathbb{Z}} c_{zN+n}(\varphi) \tilde{u}_{zN+n} \quad (\text{S52})$$

³E.g. in the libraries PyTorch and Tensorflow the resizing function provides an option of aligning the corner points of the input and output signals.

- if $\Delta \in (0, 1]$, setting $\alpha_p := c_p(\varphi)T_{p+1,p}$ and $\beta_p := c_p(\varphi)T_{p,p}$, we have

$$\tilde{v}_n \approx M \sum_{z \in \mathbb{Z}} \alpha_{zN+n-1} \tilde{u}_{zN+n-1} + \beta_{zN+n} \tilde{u}_{zN+n} \quad (\text{S53})$$

- if $\Delta \in [-1, 0)$, setting $\alpha'_p := c_p(\varphi)T_{p,p}$ and $\beta'_p := c_p(\varphi)T_{p-1,p}$, we have

$$\tilde{v}_n \approx M \sum_{z \in \mathbb{Z}} \alpha'_{zN+n} \tilde{u}_{zN+n} + \beta'_{zN+n+1} \tilde{u}_{zN+n+1} \quad (\text{S54})$$

Proof. Again, denote the DFTs of $\mathbf{u}$ and $\mathbf{v}$ by $\{\tilde{u}_m\} \in \mathbb{C}^M$ and $\{\tilde{v}_n\} \in \mathbb{C}^N$, respectively. Denote the Fourier series of φ and f by $\{c_p(\varphi)\}_{p \in \mathbb{Z}}$ and $\{c_p(f)\}_{p \in \mathbb{Z}}$, respectively.

In Sec. A, we have shown $c_p(f) = M \cdot c_p(\varphi) \cdot \tilde{u}_p$ for the interpolation process.

The next step is to link $\{\tilde{v}_n\} \in \mathbb{C}^N$ to $\{c_p(f)\}_{p \in \mathbb{Z}}$ for the discretization process. For this we need to construct a M' -periodic bridge function $g : \mathbb{R} \rightarrow \mathbb{R}$ which coincides with f on $[0, M']$, i.e. $g|_{[0, M']} = f|_{[0, M']}$, then resample $\{v_k\}$ from g by taking $v_k = g(k \frac{M'}{N})$. The Fourier series $\{c_p(g)\}_{p \in \mathbb{Z}}$ of g satisfies

$$\begin{aligned} c_p(g) &= \frac{1}{M'} \int_0^{M'} g(x) \omega_{M'}^{-px} dx \\ &= \frac{1}{M'} \int_0^{M'} f(x) \omega_{M'}^{-px} dx \\ &= \frac{1}{M'} \int_0^{M'} \left(\sum_{q \in \mathbb{Z}} c_q(f) \omega_M^{qx} \right) \omega_{M'}^{-px} dx \\ &= \frac{1}{M'} \int_0^{M'} \sum_{q \in \mathbb{Z}} c_q(f) \omega_M^{qx} \omega_{M'}^{-px} dx \\ &= \frac{1}{M'} \sum_{q \in \mathbb{Z}} \int_0^{M'} c_q(f) \omega_M^{qx} \omega_{M'}^{-px} dx \\ &= \frac{1}{M'} \sum_{q \in \mathbb{Z}} c_q(f) \int_0^{M'} e^{2\pi i x (\frac{q}{M} - \frac{p}{M'})} dx \\ &\triangleq \sum_{q \in \mathbb{Z}} c_q(f) T_{p,q} \end{aligned} \quad (*) \quad (\text{S55})$$

with

$$\begin{aligned} T_{p,q} &= \frac{1}{M'} \int_0^{M'} e^{2\pi i x (\frac{q}{M} - \frac{p}{M'})} dx \\ &= -ie^{\pi i M' (\frac{q}{M} - \frac{p}{M'})} \text{sinc} \left(\pi M' \left(\frac{q}{M} - \frac{p}{M'} \right) \right) \end{aligned} \quad (\text{S56})$$

Using $M' = M + \Delta$, we have

$$T_{p,q} = -ie^{i\pi(p - (\frac{\Delta}{M} + 1)q)} \text{sinc} \left(\pi \left(p - \left(\frac{\Delta}{M} + 1 \right) q \right) \right) \quad (\text{S57})$$

$$|T_{p,q}| = \text{sinc} \left(\pi \left(p - \left(\frac{\Delta}{M} + 1 \right) q \right) \right) \quad (\text{S58})$$

The equality at line $(*)$ applies the Fubini's theorem based on the fact that $\int_0^{M'} \sum_{q \in \mathbb{Z}} |c_q(f) \omega_M^{qx} \omega_{M'}^{-px}| dx = M' \sum_{q \in \mathbb{Z}} |c_q(f)| < +\infty$.

Eq. S55 describes each Fourier coefficient of g as a weighted sum of Fourier coefficients of f . The weights $T_{p,q}$ having a *sinc* term are negligible when $\pi M' |\frac{q}{M} - \frac{p}{M'}|$ is large. For a fixed p , we are only interested in $T_{p,q}$ whose magnitude is large. Actually, $q \mapsto |T_{p,q}|$ is maximized at $q_{max} = p \left(1 - \frac{\Delta}{M+\Delta} \right)$. We additionally limit $p \in (-M, M)$ since the Fourier coefficient $|c_p(g)|$ decreases rapidly as $|p|$ increases.

If $\Delta = 0$, $T_{p,q} = 1$ when $q = p$ and $T_{p,q} = 0$ when $q \neq p$, thus $c_p(g) = c_p(f)$. The case reduces to the aligned resampling.

If $\Delta \in (0, 1]$, $q_{max} \in [p-1, p)$. As q must be an integer, the largest two coefficients $T_{p,q}$ correspond to $q = p-1$ and $q = p$. Eq. S55 can be approximated by

$$c_p(g) \approx c_{p-1}(f) T_{p,p-1} + c_p(f) T_{p,p} \quad (\text{S59})$$

Resampling the continuous g at sampling rate $\frac{M'}{N}$ to obtain the discrete sequence $\{v_k\} \in \mathbb{R}^N$ such that $v_k = g(k \frac{M'}{N})$ is equivalent to summing up the non-overlapped intervals in length N of the Fourier transform of g [37]:

$$\begin{aligned} \tilde{v}_n &= \sum_{z \in \mathbb{Z}} c_{zN+n}(g) \\ &\approx \sum_{z \in \mathbb{Z}} c_{zN+n-1}(f) T_{zN+n, zN+n-1} + \dots \\ &\quad c_{zN+n}(f) T_{zN+n, zN+n} \\ &= M \sum_{z \in \mathbb{Z}} c_{zN+n-1}(\varphi) T_{zN+n, zN+n-1} \tilde{u}_{zN+n-1} \\ &\quad + c_{zN+n}(\varphi) T_{zN+n, zN+n} \tilde{u}_{zN+n} \end{aligned} \quad (\text{S60})$$

Setting $\alpha_p := c_p(\varphi) T_{p+1,p}$ and $\beta_p := c_p(\varphi) T_{p,p}$, we have

$$\tilde{v}_n = M \sum_{z \in \mathbb{Z}} \alpha_{zN+n-1} \tilde{u}_{zN+n-1} + \beta_{zN+n} \tilde{u}_{zN+n} \quad (\text{S61})$$

Similarly, if $\Delta \in [-1, 0)$, $q_{max} \in (p, p+1]$, Eq. S55 can be approximated by

$$c_p(g) \approx c_p(f) T_{p,p} + c_{p+1}(f) T_{p,p+1} \quad (\text{S62})$$

Setting $\alpha'_p := c_p(\varphi) T_{p,p}$ and $\beta'_p := c_p(\varphi) T_{p-1,p}$, we have

$$\tilde{v}_n = M \sum_{z \in \mathbb{Z}} \alpha'_{zN+n} \tilde{u}_{zN+n} + \beta'_{zN+n+1} \tilde{u}_{zN+n+1} \quad (\text{S63})$$

□

Theorem 6 expresses the relationship between the input DFT, the Fourier coefficients of the interpolating filter and the output DFT in the case of non-aligned resampling. In particular, let $k \in \mathbb{Z}$, one can remark that when $\Delta \neq 0$, $\tilde{v}_n$ is likely to be correlated with $\tilde{v}_{n+kM-1}$, $\tilde{v}_{n+kM}$ and $\tilde{v}_{n+M+1}$. Indeed, if $\Delta \in (0, 1]$, we have

$$\tilde{v}_n \approx M \sum_{z \in \mathbb{Z}} \alpha_{zN+n-1} \tilde{u}_{zN+n-1} + \beta_{zN+n} \tilde{u}_{zN+n} \quad (\text{S64})$$

and

$$\begin{aligned}
\tilde{v}_{n+kM-1} &\approx M \sum_{z \in \mathbb{Z}} \alpha_{zN+n+kM-2} \tilde{u}_{zN+n-2} + \dots \\
&\quad \beta_{zN+n+kM-1} \tilde{u}_{zN+n-1} \\
\tilde{v}_{n+kM} &\approx M \sum_{z \in \mathbb{Z}} \alpha_{zN+n+kM-1} \tilde{u}_{zN+n-1} + \dots \\
&\quad \beta_{zN+n+kM} \tilde{u}_{zN+n} \\
\tilde{v}_{n+kM+1} &\approx M \sum_{z \in \mathbb{Z}} \alpha_{zN+n+kM} \tilde{u}_{zN+n} + \dots \\
&\quad \beta_{zN+n+kM+1} \tilde{u}_{zN+n+1}
\end{aligned} \tag{S65}$$

which all involve the same terms $\tilde{u}_{zN+n-1}$ and $\tilde{u}_{zN+n}$ in the weighted sum.

Likewise, if $\Delta \in [-1, 0)$, we have

$$\tilde{v}_n \approx M \sum_{z \in \mathbb{Z}} \alpha'_{zN+n} \tilde{u}_{zN+n} + \beta'_{zN+n+1} \tilde{u}_{zN+n+1} \tag{S66}$$

and

$$\begin{aligned}
\tilde{v}_{n+kM-1} &\approx M \sum_{z \in \mathbb{Z}} \alpha'_{zN+n+kM-1} \tilde{u}_{zN+n-1} + \dots \\
&\quad \beta'_{zN+n+kM} \tilde{u}_{zN+n} \\
\tilde{v}_{n+kM} &\approx M \sum_{z \in \mathbb{Z}} \alpha'_{zN+n+kM} \tilde{u}_{zN+n} + \dots \\
&\quad \beta'_{zN+n+kM+1} \tilde{u}_{zN+n+1} \\
\tilde{v}_{n+kM+1} &\approx M \sum_{z \in \mathbb{Z}} \alpha'_{zN+n+kM+1} \tilde{u}_{zN+n+1} + \dots \\
&\quad \beta'_{zN+n+kM+2} \tilde{u}_{zN+n+2}
\end{aligned} \tag{S67}$$

which all involve the same terms $\tilde{u}_{zN+n}$ and $\tilde{u}_{zN+n+1}$ in the weighted sum.

Therefore, in this case of resampling a signal of M samples into N samples with slightly non-aligned resampling interval, two frequency components separated by $kM - 1$, kM or $kM + 1$ components in the spectrum of the resampled signal are likely to be correlated. Moreover, as $|k|$ increases, the correlation vanishes rapidly. This corresponds to the observation in Fig. 2 that the significant distances d related to anomalous spectral correlations belong to $\llbracket M - 1(\bmod N), M + 1(\bmod N) \rrbracket \cup \llbracket -M - 1(\bmod N), -M + 1(\bmod N) \rrbracket$.

APPENDIX F

IMPLEMENTATION AND COMPUTATIONAL COST

The implementation of the *a contrario* detection method in Sec. III-A is detailed in Alg. 1, where the most time-consuming part is the calculation of the correlations between $\mathcal{O}(L \times N)$ patch pairs. The time complexity of computing one correlation is $\mathcal{O}(l \times l)$. Thus, the total time complexity is $\mathcal{O}(L \times N \times l \times l) = \mathcal{O}(L^3)$, where $N \times l \times l$ is the number of pixels in the order of magnitude of L^2 . To accelerate the computation, the correlation of $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$ is computed in the form of

$$\text{corr}(\mathbf{x}, \mathbf{y}) = \frac{\sum x_i \bar{y}_i - \frac{1}{n} \sum x_i \sum \bar{y}_i}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2} \tag{S68}$$

where we pre-compute the sums $\sum x_i$ and $\sum y_i$ and the norms $\|\mathbf{x}\|_2$ and $\|\mathbf{y}\|_2$ of all the used patches, and only compute the inner product $\sum x_i \bar{y}_i$ of two patches and reuse the precomputed norms and sums at line 6 in Alg. 1. The calculation of the sums and the norms of the patches can be accelerated by the image integral technique [16] to reduce the time complexity from $\mathcal{O}(L \times N \times l^2) = \mathcal{O}(L^3)$ to $\mathcal{O}(L^2)$.

Algorithm 1: Calculation of NFAs for detecting anomalous local correlations in the spectrum.

Input : $\mathcal{P}$, a list of N non-overlapped spectrum patches of size $l \times l$
Input : L , the length of the side of detection
Input : r , the range of local maximum for validating a correlation
Output : NFAs for distances in $\llbracket 0, L - 1 \rrbracket$

```

1  $\rho \leftarrow \text{zeros}(L, N)$ 
2 for  $d \in \llbracket r + 1, L - r - 1 \rrbracket$  do
3   for  $i \in \llbracket 0, N - 1 \rrbracket$  do
4      $P \leftarrow \mathcal{P}(i)$ 
5      $Q \leftarrow$  a patch  $d$  components apart from  $P$ 
6      $\rho(d, i) \leftarrow \text{corr}(P, Q)$ , see Eq. 15
7   end
8 end
9  $\text{NFA} \leftarrow$  a list of  $L$  values initialized by  $+\infty$ 
10 for  $d \in \llbracket r + 1, L - r - 1 \rrbracket$  do
11    $k \leftarrow 0$ 
12   for  $i \in \llbracket 0, N - 1 \rrbracket$  do
13     if  $\rho(d, i) = \max\{\rho(d', i) \mid d' \in \llbracket d - r, d + r \rrbracket\}$ 
14       then
15          $k \leftarrow k + 1$ 
16       end
17    $\text{NFA}(d) \leftarrow (L - 2r - 1) \cdot \mathbf{B}\left(k, N, \frac{1}{2r+1}\right)$ ,  $\mathbf{B}$  is the
      tail of binomial law
18 end
19 return NFA

```

Regarding the preprocessing step, the rank transform takes a local neighborhood of size 7×7 , and the TV denoiser is detailed in [15] with parameter $\lambda = 1$.

The computational cost was evaluated on Apple Mac M1 Pro with CPU at 3.2GHz and memory bandwidth at 200GB/s. The

method was implemented in Python. On a 600×600 image, the average execution time of the *a contrario* detection method along one axis was 0.66s. The preprocessing took 0.40s using the rank transform and 0.87s using the TV denoiser.